

Excitation-Aware On-Track System Identification at Full Scale: A Simulation Study

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Abstract—On-track system identification promises tire models learned from ordinary racing laps rather than dedicated manoeuvres. We study one such residual-learning pipeline at full vehicle scale, in a simulator publishing per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform it converges on a tire with less than half the vehicle's grip ceiling and most Pacejka coefficients railed: not a solver or tuning failure but a loss of excitation created by the pipeline's own virtual-data step. The friction the synthetic rollout can demand, $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}} / (gL)$, is a property of the rollout configuration and not of the tire, as a grid on a known tire confirms. Two further defects share that source: the fit's objective depends on the coefficients only through the product BCD , so the peak factor is not recoverable from the rollout at any excitation, and the loop feeding each fit its own previous answer is not a contraction. We correct all three (rollout speed from the measured lap, sweep bounded to the observed steering, peak factor measured off the trace and pinned, update damped into a relaxed fixed-point iteration) and gate the controller interface on derived physical quantities, not coefficient bounds. Over three timed laps on a surface 33 % below the one its prior was measured on, no coefficient rails, axle cornering stiffness is recovered to 3 % and the peak factor to 11 %, against a railed peak factor and a 125 % stiffness error for the inherited configuration. All results are in simulation, one run per configuration.

Index Terms—Autonomous vehicles, system identification, parameter estimation, vehicle dynamics, tires, predictive control, road vehicles, simulation.

I. INTRODUCTION

MODEL-BASED controllers for autonomous racing derive their performance from the fidelity of the vehicle model they carry [1], [2], [3]. In the lateral channel that fidelity is dominated by the tire, simultaneously the physical bound on manoeuvrability and the least accessible component to measure: its behaviour depends on surface, temperature, wear and load in ways that change between sessions [4], [5].

Classical characterisation resolves this by taking the vehicle off the track. Steady-state circular or ramp-steer manoeuvres in a large open space yield clean data, at a logistical cost a race weekend rarely affords [6]. Identifying on track removes the logistics and brings back transients, noise and a racing line that is a poor excitation signal for a nonlinear model; nonlinear least squares applied directly to lap data is brittle in exactly this regime [6].

Dikici *et al.* [6] propose the hybrid remedy that is the starting point of this paper. A small residual network is trained

on 30 s of ordinary lap data to predict the one-step error of a nominal single-track model. The corrected model is rolled out through a *synthetic*, quasi-steady steering sweep, and Pacejka coefficients are fitted to that sweep. Because the sweep is generated rather than driven it can be made steady and noise-free, and because the residual network only has to interpolate within its training distribution, its generalisation weakness stays contained. The procedure iterates, each fit becoming the next nominal model. On a 1:10 platform it reports a $3.3\times$ lower one-step RMSE than nonlinear least squares under noise, from 30 s of driving [6].

A. Problem statement

This paper asks what the pipeline identifies when it is moved from a 3.7 kg, 0.32 m-wheelbase platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase, 16° of road-wheel lock, following a reference speed profile of 9.6–25.8 m/s and achieving 9.2–26.5 m/s in the identification data. The question is identifiability, not parameter transfer, and two scale-dependent mechanisms, both latent at 1:10, break the estimate:

- 1) **Excitation collapses inside the virtual-data step.** The Pacejka fit never sees the recorded lap. It sees the synthetic rollout, whose reachable friction $\mu_{\text{reach}} \approx v^2 \delta_{\text{sweep}} / (gL)$ is a property of the rollout configuration and not of the tire, and a rollout speed inherited from the scaled platform caps it at 0.43 here. Below the tire's peak the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$, only the product BCD is identifiable, and the bounded optimiser resolves the remaining freedom on a box edge [7], [4].
- 2) **Lateral demand scales as v^2 , so a bad grip estimate is silent until it is catastrophic.** Driven through the same controller on an earlier, faster reference trajectory, one identified set with a peak friction coefficient of 0.40 tracked at 0.16 m of cross-track error at 11.1 m/s, 0.43 m at 15 m/s and 41.89 m at 27 m/s. The scaled platform never reaches the speed at which the error becomes observable. These figures come from an offline closed-loop harness rather than a controlled simulator run and are reported as such throughout (Section VII-A).

One difference works in our favour. Because the study runs in the CARLA simulator [8] through a purpose-built ROS 2 interface, the simulator's own per-wheel telemetry (slip angle, normal load and lateral force) is available as ground truth, so

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an on-track identifier can be scored against the forces the plant actually generated.

B. Contributions

- 1) **An identifiability analysis of the virtual-data step** (Section V-B), giving the closed-form reachable-friction bound above and an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns whenever the configuration cannot reach the prior's peak. A grid over rollout speed, sweep amplitude and true peak factor on a known tire verifies it (Section VI-B): below the tire's own peak the fit returns the excitation rather than the tire, at every grip level tested and independently of the solver. This corrects the baseline method [6] at any scale, and stays latent at 1:10 because a 2–4 m/s rollout is representative there and is not here.
- 2) **Two structural properties of the loop, and the corrections they imply** (Section V-D). Because stage 3 reconstructs its forces from the rollout's own quasi-steady yaw balance, it observes nothing but the lateral acceleration the previous coefficient set produced, so its residual depends on B , C and D only through their product: the peak factor is not recoverable from the rollout *at any excitation*, a stronger statement than the low-slip degeneracy, and must instead be measured on the recorded trace and held. And because each fit is handed the trajectory its own previous answer generated, the loop is a positive feedback path rather than a contraction, and iterating it undamped degrades the estimate monotonically. Stage 4 becomes a relaxed fixed-point iteration at $\beta = 0.20$; the $\beta = 0$ control, which performs no identification at all, is the worst setting of the sweep, so the iterations do carry information.
- 3) **The corrections the bound implies**, on both sides of the fit. On the rollout side (Section V-C), the sweep is bounded at $1.5\times$ the 99th percentile of the observed steering, so the fit cannot read the residual network's extrapolation as grip, and the Tikhonov target is frozen at the configured prior while the optimiser's start point tracks the iterations; together those remove a drift that took D from the 1.009/1.002 front/rear prior to 1.5402 and 1.8721 over six iterations. On the measurement side (Section V-F), slip angles are built from the *achieved* road-wheel angle rather than the command, mass and wheelbase from measured static wheel loads rather than from a configuration file, the friction warm start from the *curvature* of the axle force curve rather than a median over a low-slip mask, and the dynamics residuals of the physics-informed objective are normalised to the state-increment units of the data term, which the published form leaves a factor $2.3e5$ apart.
- 4) **Admissibility gates on derived physical quantities** (Section VII) at the model-to-controller boundary: axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the trajectory's lateral demand, rather than per-coefficient box checks, which cannot

detect a degenerate fit or a pairwise-oversteering axle set.

- 5) **A simulation architecture that supplies the ground truth, and a closed-loop evaluation on it** (Sections IV-B–IV-E and VI). CARLA_ASU_BRIDGE exposes the plant's own per-wheel forces, slip angles and loads as ROS 2 topics alongside a conventional autonomy interface. The evaluation runs on a static 0.70 friction surface, deliberately not the 1.05 the configured tire prior was measured on, so that a pipeline unable to move off its prior has nowhere to hide (Section VI-A4). Over three timed laps the corrected pipeline identifies axle cornering stiffness to 3.2 % front and 1.4 % rear and tracks the peak factor to 11 %, against an inherited configuration that rails D on its 2.0 bound in every cycle and overstates stiffness by 118–132 %. Axle force RMSE against the plant's telemetry is $12.1\times$ and $10.5\times$ lower, and MPC lateral tracking error 32 % lower, at unchanged solver cost.

All results are obtained in simulation, and Section VIII states which conclusions are expected to transfer to hardware.

II. RELATED WORK AND POSITION

A. Tire models and Magic Formula identification

The Magic Formula of Bakker, Nyborg and Pacejka [9] and its consolidation [10], [4] remains the standard empirical description of steady-state tire force generation, and it is the model identified here; its lateral form and the two properties this paper relies on are stated in Section III. Classical characterisation obtains the coefficients from steady-state manoeuvres in a controlled space [6], [5]: clean data at high logistical cost. Direct fitting to measured contact-patch forces is the most informative variant and needs a force-measuring hub or a simulator's own wheel telemetry; here it is used only as a *reference* against which the on-track identifier is scored, never inside the identification loop. Between the two sits operator-guided identification from track or rig data, of which TRIP-ID [11] is representative. The pipeline studied here targets the same output with no operator and no dedicated manoeuvre, which is exactly what puts its excitation, and therefore its identifiability, outside the engineer's control.

B. On-track and data-driven identification

On-track methods trade excitation quality for operational simplicity. Nonlinear least squares on lap data is the classical baseline and is brittle under noise [6]. The recursive alternative is a dual or joint filter carrying states and parameters together, of which Wenzel *et al.* [12] is the canonical vehicle instance and Sierra *et al.* [13] the cornering-stiffness variant. It is subject to the same excitation condition this paper is about: a filter run on a smooth racing line has as little information about the tire's peak as a batch fit of the same data, and merely expresses that as a parameter covariance that never shrinks rather than as a coefficient on a bound. Dedicated friction estimators make the excitation dependence explicit [14], [15], [16], [17], and Hsu *et al.* [18] is the closest prior art to the warm start of Section V-F3: they recover

the friction limit below the peak from the *curvature* of the force response, read through the pneumatic trail in steering torque. Our estimator uses the same observation on a different signal, the axle force curve reconstructed from the IMU and odometry, with no steering-torque sensor and no assumption about the steering system. It is then gated, applied to both axles from the axle the manoeuvre identifies, and pinned for the co-identification loop — a property of the controller interface and of what that loop can identify (Section V-D1) rather than of the estimator. Gaussian-process approaches learn the mismatch of a nominal model, for identification and inside the controller alike [19], [20], [2]; they are accurate, but carry a continuing computational cost and, as [6] observes, are usually evaluated starting from an already accurate physical model. Neural residuals replace the Gaussian process with a cheaper regressor at the cost of generalisation guarantees.

Deep Dynamics [21] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer clamping them into nominal ranges. Our admissibility gates share the motivation but differ in placement. We report as a substantive finding that per-coefficient box constraints are *insufficient*: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 14.5 m/s. We therefore gate on derived quantities at the model-to-controller interface and treat a coefficient resting on a bound as a diagnostic of non-identifiability rather than a successful clamp.

The hybrid residual-plus-virtual-data structure of [6] is the pipeline we extend. Its key idea, to query the learned model only where it interpolates and extract physically parameterised coefficients from it, is retained unchanged. What we correct is the mechanism deciding *where* the learned model is queried.

C. Overlap with concurrent work

Wu *et al.* [22] extend the same baseline concurrently and independently, introducing a vision-based friction prior, an S4 structured state-space residual in place of the MLP, and Nelder–Mead for the iterative Pacejka extraction. The pipeline described here independently contains an S4D temporal residual [23], [24], a friction warm-start, and Nelder–Mead and differential-evolution solver options, so we claim none of the three as contributions. Three differences are technical rather than chronological. Our friction prior is proprioceptive, so it needs no camera and no textured-surface assumption. It is read from the *curvature* of the axle force curve by a gated brush-model fit [25], [16] rather than from utilised friction, which on a gentle lap measures 0.044 median on this vehicle and is only a lower bound (Section V-F3d). And it is *pinned* rather than used as an initialisation, because Section V-D1 shows the rollout fit cannot separate the peak factor from the cornering stiffness at any excitation, so a prior on D the fit is free to move is overwritten within one cycle. Neither prior work is the fix: the degeneracy is located in the rollout excitation, not the initialisation, and to our knowledge the reachable-friction bound of Section V-B and its consequence for the virtual-data step have not been stated before.

D. Simulation architectures for identification and validation

CARLA [8] is the standard open simulator for autonomous-driving research and supplies the sensor and scenario infrastructure used here. Its native dynamics come from the underlying PhysX wheeled-vehicle model [26], a saturating friction-and-stiffness tire law (Section IV-C) rather than an empirical Magic Formula fit to measured data, which is why this study treats the simulator as a well-instrumented *plant* to be identified and not as a source of ground-truth Pacejka coefficients. Recent work addresses the fidelity gap directly: R-CARLA [27] makes the dynamics model interchangeable while retaining CARLA’s sensor simulation, and Sagmeister *et al.* [28] quantify how far a vehicle model can be simplified before conclusions about a trajectory-following controller stop transferring, finding, relevantly here, that sensitivity is largest near the acceleration limits. CARLA has also hosted complete Formula Student Driverless stacks with ROS 2 interfaces [29]. The architecture of Section IV sits in this line, distinguished by exposing the simulator’s own per-wheel slip, load and force telemetry on a simulation-time-stamped interface, so that an identification method can be scored against the forces the plant produced rather than a secondary model. Scaled platforms, F1TENTH [30] and the ForzaETH Race Stack [3], provide the software context of the baseline; full-scale Formula Student Driverless systems [5] the vehicle class targeted here.

E. Downstream consumer and gap addressed

The identified model is consumed by a nonlinear MPC path-tracking controller built on *acados* [31] with partial-condensing HPIPM [32], following standard practice in racing MPC [1], [2], tracking a minimum-curvature raceline generated in the manner of Heilmeier *et al.* [33]. We claim no contribution to the controller; it appears only as the consumer that defines what an admissible identification is (Section VII).

The gap, then. The baseline [6] establishes that a residual network plus virtual steady-state data recovers a Pacejka model from a lap at 1:10 scale, and concurrent work [22] improves its residual architecture and initialisation. Neither addresses the condition under which the virtual data itself carries information about the tire’s peak, the binding constraint at full scale, and neither treats the admissibility of the identified model for its downstream controller as an explicit, physically derived contract. This paper addresses both, on a platform where the answer can be checked against the plant’s own forces.

III. IDENTIFICATION PROBLEM

A. Model structure

The identification model is the dynamic single-track model of the baseline, with mass m , yaw inertia I_z and centre-of-gravity distances l_f , l_r , $L = l_f + l_r$. Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \frac{1}{m}(F_{y,r} + F_{y,f} \cos \delta - mv_x \omega), \quad (1)$$

$$\dot{\omega} = \frac{1}{I_z}(F_{y,f} l_f \cos \delta - F_{y,r} l_r), \quad (2)$$

with slip angles

$$\alpha_f = \delta - \arctan\left(\frac{v_y + l_f \omega}{v_x}\right), \quad \alpha_r = -\arctan\left(\frac{v_y - l_r \omega}{v_x}\right). \quad (3)$$

Axle forces follow the lateral Magic Formula

$$F_y = F_z D \sin(C \arctan(B\alpha - E(B\alpha - \arctan B\alpha))), \quad (4)$$

with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The identification state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$, and the nominal one-step map is $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. The baseline discretises (1)–(2) by explicit Euler at T_s . We keep Euler as the default and expose second- and fourth-order Runge–Kutta as options, because the full-scale vehicle’s lateral mode is stiffer relative to T_s than the scaled platform’s.

B. Objective and derived quantities

The objective is to recover Φ_p from a single 30 s segment of ordinary lap data, without dedicated manoeuvres, such that the resulting model is both predictive and *admissible* for the controller that consumes it. Admissibility is expressed on derived quantities rather than on coefficients. The linearised axle cornering stiffness is the slope of (4) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\text{MF}} D_i, \quad i \in \{f, r\}, \quad (5)$$

(writing C^{MF} for the shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\max} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \quad (6)$$

The linear model built from (5) is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\text{crit}} = L \sqrt{C_f C_r / (m(l_f C_f - l_r C_r))}$ [34].

C. Practical identifiability under limited excitation

Two properties of (4) are used throughout. First, its slope at zero slip is $F_z BCD$, so the cornering stiffness depends on the coefficients only through that product, and E drops out. Second, and decisively, if the data never approaches the peak, (4) linearises to $F_y \approx F_z (BCD)\alpha$ and B, C, D become mutually unidentifiable. Given data that does reach the peak, (4) is *structurally* identifiable in the sense of Bellman and Åström [35], the map from coefficients to noise-free outputs being injective. What fails here is *practical* (a-posteriori) identifiability [36]: under a specific excitation the likelihood is flat along the BCD direction, so the parameters are not recoverable from the data at hand even though the structure admits them. That is the standard persistent-excitation condition [7], [37] written on this model, and we claim neither it nor the structural result as novel. Our claim, demonstrated in Section V-B, is that the baseline pipeline’s virtual-data step can silently violate that condition, and that a bounded optimiser then resolves the remaining freedom by sliding onto a box edge. The violation surfaces as a converged fit rather than a failure.

D. Baseline pipeline

The baseline proceeds in four stages, repeated for N_{it} iterations [6]:

- 1) **Residual learning.** A small network f_{NN} is trained on the recorded lap to predict the one-step error $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ from $[v_x, v_y, \omega, \delta]_k$, and the corrected model is $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1} + e_k$.
- 2) **Virtual steady-state generation.** The corrected model is integrated open-loop at constant longitudinal speed while the steering angle is ramped from 0 to δ_{sweep} , producing a synthetic quasi-steady dataset.
- 3) **Pacejka fit.** Assuming $\dot{v}_y = \dot{\omega} = 0$ on the synthetic data, axle forces are recovered from the yaw balance,

$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \quad F_{y,f} = \frac{m l_r v_x \omega}{L \cos \delta}, \quad (7)$$

slip angles from (3), and Φ_p is fitted to $(\alpha_i, F_{y,i})$ by bounded least squares.

- 4) **Re-nomination.** The fitted Φ_p becomes the next iteration’s nominal model and the residual network is reinitialised.

The point that drives the rest of this paper is that stage 3 *never sees the recorded lap*. It sees the output of stage 2. The recorded lap enters only through the residual network, so the excitation available to the fit is set by the rollout configuration and not by how the vehicle was driven.

IV. SIMULATION ARCHITECTURE AND DATA PIPELINE

The pipeline is exercised end to end inside a single architecture: a Formula Student vehicle model in CARLA acting as the plant, a C++ ROS 2 communication layer that drives it and republishes its state, and an identification stack that consumes that state.

A. Choice of simulator

CARLA [8] was selected against three requirements the identification problem imposes, the first two of which are what make the identifiability claims of Section V checkable rather than merely arguable. **Per-wheel ground truth:** the claim to be checked is that the recovered Magic Formula reproduces the lateral force the plant produced, so the simulator must expose per-wheel slip angle, normal load and tire force, not only chassis states; CARLA’s PhysX solver computes these internally and its client API exports them, which is what makes the tire-force feedback channel of Section IV-D possible. **Deterministic, simulation-time-stamped execution:** persistent-excitation and residual-learning arguments are only checkable if every sample carries a consistent clock, and CARLA’s synchronous mode with a fixed physics step (0.033 s here) decouples data timing from wall-clock load (Appendix, Section A7). **Full-scale vehicle and racing-track support:** CARLA accepts custom vehicle assets, here the Formula AI vehicle on the *silverstone* map, and offers a client the bridge links against directly. Surveys of open self-driving simulators place CARLA at the state of the art for end-to-end stack testing on these grounds [38].

TABLE I
VEHICLE CONFIGURATION AND MEASURED PLANT PROPERTIES.

Quantity	Value	Unit
Body mass (configured)	240.0	kg
Static wheel load (total)	2644.5	N
Total (kerb) mass m^a	269.6	kg
Unsprung mass ^b	29.6	kg
Front/rear static split	51.2 / 48.8	%
Wheelbase L	1.5334	m
l_f / l_r	0.7381 / 0.7953	m
Yaw inertia I_z^c	51.1	kg m ²
Wheel radius	0.25	m
Max road-wheel angle	16.0	°
Configured wheel friction ^d	1.0	—
Road-surface factor	0.70	—
Effective friction ^e	0.70	—
Realised peak axle friction	0.71	—
Drag coefficient	0.426	—

Configured values are set in the bridge's configuration; measured values are obtained from the simulator's per-wheel telemetry. ^aTotal static wheel load divided by $g = 9.81 \text{ m/s}^2$ ($2644.5/9.81 = 269.6$). This is the mass every equation in this paper uses, not the configured body mass. ^bTotal minus body mass; not measured independently. ^cAssumed; see Section A. ^dA scenario parameter, not a vehicle property: the runs of Section VI set it to 1.0; the dedicated limit-excitation run that produced the tire prior of Section VI-A4 used 1.5. ^eConfigured wheel friction multiplied by the road surface's own coefficient, as reported per wheel by `feedback/tire_forces`. It is the effective value, not the configured one, that the identifier is scored against (Section V-F3d).

B. Plant: Formula A1 vehicle in CARLA

The plant is the driverless Formula Student-class vehicle model, instantiated in CARLA [8] (server 0.9.16, Unreal Engine 4.26) on the `silverstone` map: a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle whose lateral forces are generated by the PhysX wheeled-vehicle tire model [26] under a configured friction coefficient and lateral stiffness, not by a Magic Formula. The simulator therefore has no "true" Pacejka coefficients to recover. Instead the lateral forces PhysX itself computes at each wheel are taken as ground truth, and the identified Magic Formula is validated against those forces (Section V-F).

C. The plant's tire curve, in closed form

The identification target is worth stating exactly, because it is knowable here. PhysX computes the lateral force of each wheel with `PxVehicleComputeTireForceDefault` [26]; at zero longitudinal slip and zero camber that expression collapses to

$$F_y(\alpha) = \text{sgn}(\alpha) \mu F_z S_1(K), \quad K = \frac{C_{\text{lat}} |\tan \alpha|}{\mu F_z}, \quad (8)$$

with $C_{\text{lat}} = F_z^{\text{rest}} k_y S_1(3\bar{F}_z/k_x)$, $\bar{F}_z = F_z/F_z^{\text{rest}}$ the normalised load, (k_y, k_x) the wheel's lateral-stiffness parameters, and $S_1(K) = \min(1, K - K^2/3 + K^3/27)$ the engine's smoothing function.

Two properties of (8) shape everything downstream. It saturates at μF_z , first reached at $K = 3$, and it *never falls off* past that peak, so a Magic Formula, whose E and post-peak decay exist precisely to describe a falling branch, can match it up to the peak and no further. Its scale is set entirely

TABLE II
THE PLANT'S LATERAL TIRE CURVE: THE PARAMETERS OF (8), READ FROM THE SIMULATOR'S OWN WHEEL CONFIGURATION AND TELEMETRY.

Quantity	Value	Unit
Effective friction μ	0.70	—
Lateral stiffness k_y	17.0	/rad
Max normalised load k_x	2.0	—
Initial slope C_{lat} at $\bar{F}_z = 1$	$14.875 F_z^{\text{rest}}$	N/rad
front axle, $F_{z,f} = 1373 \text{ N}$	20.4	kN/rad
rear axle, $F_{z,r} = 1274 \text{ N}$	19.0	kN/rad
Peak reached at, $\bar{F}_z = 1, \mu = 0.70$	8.0	°

The initial slope is set by the *rest* load and by the normalised load through S_1 , not by F_z alone: $C_{\text{lat}} = k_y S_1(3\bar{F}_z/k_x) F_z^{\text{rest}}$ evaluates to $14.875 F_z^{\text{rest}}$ at $\bar{F}_z = 1$ and falls monotonically with $\bar{F}_z$.

Every axle slope quoted in this paper is therefore taken at $\bar{F}_z = 1$ and at the static axle loads listed here (the geometric split $l_r/L, l_f/L$ of the total wheel load; see Section A, A2, for its 1.4% disagreement with the measured load split of Table I).

The peak angle follows from $K = 3$ in (8), i.e. $\tan \alpha_{\text{peak}} = 3\mu/(k_y S_1(3\bar{F}_z/k_x))$, and therefore moves with μ : 8.0° at the effective $\mu = 0.70$ of the runs reported here, 12.0° at the $\mu = 1.05$ of the prior-fitting run. The initial slope does not move with μ ; the *effective* slope fitted over a given run's load range does, and falls below the $\bar{F}_z = 1$ value tabulated here.

by μ and C_{lat} , both of which the simulator publishes, so the curve of Table II is read off the plant rather than fitted to it.

That the closed form is the right one was checked rather than assumed. On 1804 ground-truth wheel samples from a live lap ($P_{99}(|\alpha|) = 3.66^\circ$), each predicted from its own measured α , F_z and μ , (8) returns $R^2 = 0.99$ with 16.1 N RMSE on the front axle and 14.7 N on the rear, at biases of +2.0 N and +0.8 N. Section VI-D uses this curve to separate the error of the identified fit from the error of representing (8) in the form (4) that the controller requires.

D. CARLA_ASU_BRIDGE: interfaces and telemetry

CARLA_ASU_BRIDGE [39] is a C++ ROS 2 lifecycle node linked directly against CARLA's client library (Fig. 1). It spawns and configures the ego vehicle, drives its inputs, and republishes simulator state as standard ROS 2 topics; all identification traffic crosses this interface. The command path is a single `ackermann_msgs/AckermannDriveStamped` on `/drive`, forwarded to CARLA's native Ackermann controller; no client-side steering PID is present, so the achieved/commanded ratio of Table I is a property of the vehicle rather than of a tuning choice in the bridge. On the feedback path, one property is specific to this architecture and the rest of this paper rests on it: the simulator's per-wheel slip angle, normal load and lateral force are republished on the same simulation clock as the odometry the identifier consumes, so an identified tire model can be scored against the forces the plant actually generated rather than a second model. Appendix B lists the full interface, the body-frame and sign conventions at the simulator-ROS 2 boundary, and the six extensions made to the bridge for this study.

E. Data acquisition and preprocessing

A model-free pure-pursuit controller drives the vehicle around the reference raceline. It requires no tire knowledge beyond the wheelbase and is therefore a valid bootstrap, as

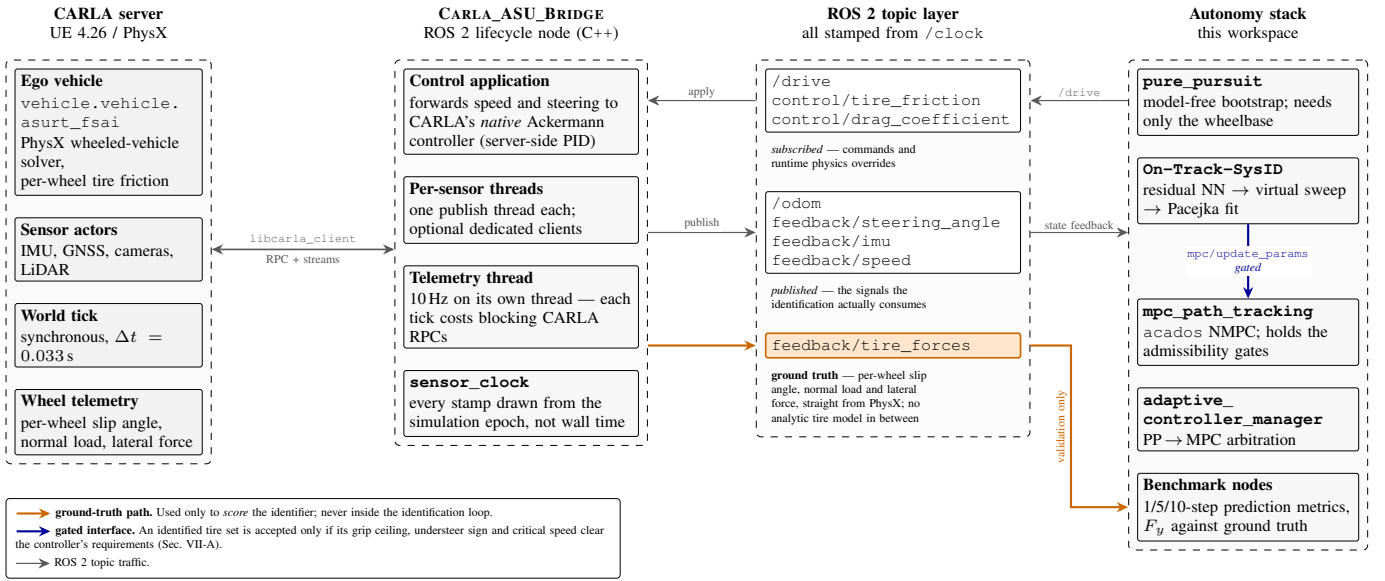


Fig. 1. Architecture of the CARLA_ASU_BRIDGE ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel telemetry on ROS 2 topics stamped from the simulation clock. The *feedback/tire_forces* path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

in the baseline [6], [3]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer by a 50 Hz timer for 30 s, gated on $v_x > 1$ m/s. The timer runs on the simulation clock, which advances only at the server's 0.033 s step, so the achieved acquisition rate is the step rate, not the timer rate.

Two mismatches follow, with different consequences. The acquisition timer asks for a sample every 0.020 s, 65 % more often than the server produces one, so a fraction of consecutive buffer entries are repeated states and the one-step residual target $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ carries a zero-order-hold artefact on those pairs. Separately, the one-step model of Section III-A is discretised at $T_s = 0.035$ s, 6 % longer than the physics step. T_s is empirical, matching neither clock, and is carried because the identification proved insensitive to it over the range tested: at $T_s = 0.020$ s and $T_s = 0.035$ s the corrected configuration returns the *same* last accepted coefficient set to four decimal places ($B, C, D, E = 7.076, 1.346, 1.009, -2.000$ front and $7.873, 1.383, 1.002, -1.024$ rear), while in the inherited configuration the coefficients move (B_f 4.72→4.80, E_f -2.44→-2.26) with D on its 2.0 bound at both settings — so the box-edge signature is not an artefact of the mismatch either. Results are reported at $T_s = 0.035$ s. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$ at unchanged v_x , which doubles the data and removes the cornering bias of a circuit. Re-identification retriggers every 30 s by default.

F. How the architecture closes the loop

Acquisition uses */odom*, *feedback/steering_angle* and */drive*, all stamped on the simulation clock. *Preprocessing* and *fitting* run inside the identification node against that buffer. *Validation* draws on *feedback/tire_forces* and *feedback/imu*, neither of which enters the identification

loop, so the estimator is scored against a channel it did not observe. *Deployment* pushes the fitted coefficients over a ROS 2 service to the controller behind the gates of Section VII. Only the last stage leaves the architecture, which is what lets Section VI attribute a closed-loop difference to the identified model rather than to the interface carrying it.

V. IDENTIFICATION METHODOLOGY

Figure 2 summarises the loop and Algorithm 1 states it in the form needed to reimplement it; lines marked $\triangleright$ are this paper's changes to the inherited pipeline. Sections V-B–V-C carry the central methodological contribution, the two changes to the virtual-data step.

A. Residual model

The residual carries whatever the nominal model (1)–(4) does not represent. How that map is parameterised is a modelling decision: known vehicle physics shapes its input space and, optionally, its objective (Section V-A1), so that the network's small capacity is spent on what is genuinely unmodelled rather than on rediscovering structure (3) and (1)–(2) already state exactly.

The default residual is the baseline multilayer perceptron: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU, 58 parameters, full-batch Adam [40] at 5×10^{-4} on an MSE objective, in PyTorch [41]. Because the full-scale residual has different structure from the scaled platform's, the architecture is selectable: baseline, wide (16 hidden units, 114 parameters, weight decay), physics_inputs (74 parameters), ensemble (three averaged members, 174 parameters), and s4 (102–110 parameters). The results below use the baseline MSE objective, so that the identifiability statements of Sections V-B–V-C are not confounded with a change of residual.

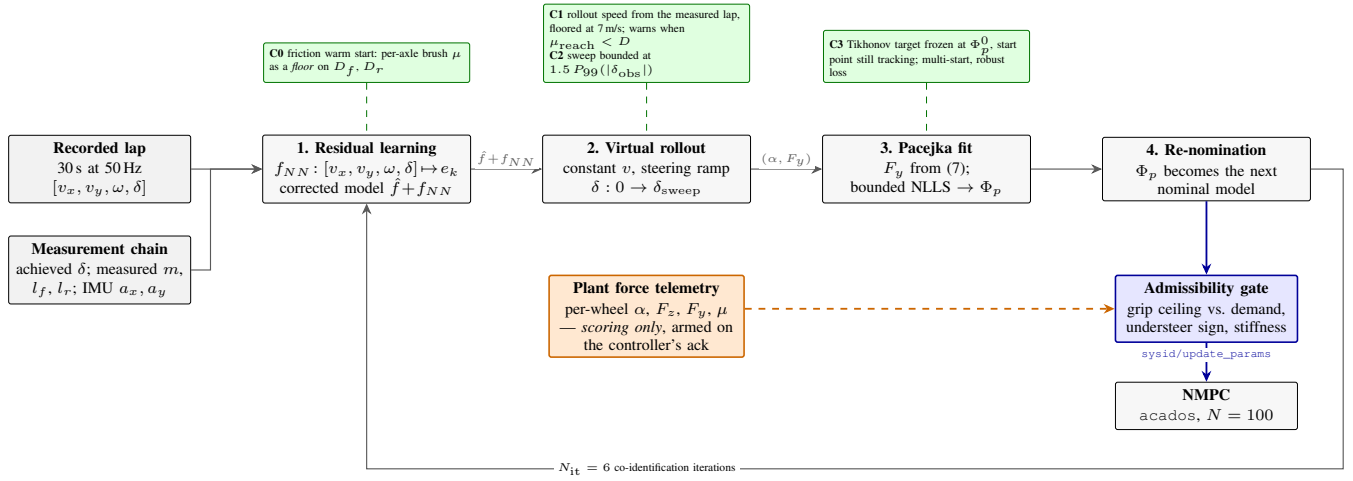


Fig. 2. The identification pipeline. Recorded lap data trains the residual network, the corrected model is rolled out through a synthetic steering sweep, Pacejka coefficients are fitted to that sweep, and the fit becomes the next nominal model under relaxation. Green marks this paper's changes to the inherited loop: the friction warm start, which measures D off the recorded trace and pins it for the loop (Section V-F3), the excitation-aware rollout speed and the extrapolation-bounded sweep (Sections V-B–V-C), the frozen regularisation target of (12), and the relaxed coefficient update of (14). Blue is the gated model interface to the controller (Section VII); orange is the plant's own per-wheel force telemetry, which scores the result, is armed only once the controller has acknowledged the submitted model (Section V-G), and is never admitted into the loop.

```

in: buffer  $\mathcal{D} = \{v_x, v_y, \omega, \delta^{\text{ach}}\}_{1:K}$  (30 s), prior  $\Phi_p^0$ , bounds  $[\Phi_p^-, \Phi_p^+]$ 
out:  $\Phi_p$ , or UNIDENTIFIED

1:  $\delta \leftarrow$  achieved road-wheel angle, command on staleness
2:  $m, l_f, l_r \leftarrow$  measured static wheel loads
3:  $\hat{\mu}_f \leftarrow \text{BRUSHFIT}(\mathcal{D})$ , front axle, released only if gated
4: if released then  $D_f^0, D_r^0 \leftarrow \hat{\mu}_f$  (two-sided, both axles)
5:  $D$  held fixed for the whole loop, released or not
6:  $\Phi_p^{\text{tik}} \leftarrow \Phi_p^0$  (frozen for all iterations)
7:  $v \leftarrow \text{clip}(\bar{v}_x(\mathcal{D}), v_{\min}, \infty)$ 
8:  $\delta_{\text{sweep}} \leftarrow \text{clip}(1.5 P_{99}(|\delta|), \delta_{\min}, \delta_{\max})$ 
9:  $\mu_{\text{reach}} \leftarrow v^2 \delta_{\text{sweep}} / (g L)$ ; warn if  $\mu_{\text{reach}} < \max_a D_a^0$ 
10:  $\Phi_p \leftarrow \Phi_p^0$ 
11: for  $i = 1$  to  $N_{\text{it}}$  do
12:    $f_{NN} \leftarrow \text{TRAINRESIDUAL}(\mathcal{D}, \Phi_p)$  (re-initialised each  $i$ )
13:    $S \leftarrow \text{ROLLOUT}(\Phi_p, f_{NN}, v, \delta_{\text{sweep}})$ 
14:    $\Phi_p \leftarrow \arg \min_{B,C,E} \|F_y(\Phi_p; S) - \hat{F}_y(S)\|^2 + \lambda \|\Phi_p - \Phi_p^{\text{tik}}\|^2$ ,
15:   subject to  $\Phi_p \in [\Phi_p^-, \Phi_p^+]$ , started from  $\Phi_p$  of  $i - 1$ 
16:    $\Phi_p \leftarrow (1 - \beta) \Phi_p + \beta \Phi_p$  ( $\beta = 0.20$ )
17:   if any  $\Phi_p$  component on a bound then report the degeneracy
18:   if not  $\text{GATES}(\Phi_p)$  then return UNIDENTIFIED
19: submit  $\Phi_p$ ; keep the previous model until acknowledged

```

: GATES: axle stiffness sign and magnitude; $l_f C_f < l_r C_r$ and v_{crit} outside the speed range; grip ceiling $\geq$ the trajectory's lateral demand (Section VII).

Algorithm 1: Corrected on-track identification cycle. $\triangleright$ marks a change to the inherited pipeline [6].

1) *Physics-augmented inputs and objective*: The input space is augmented with the axle slip angles of (3): since the tire law (4) is a function of (α, F_z) and of nothing else in $\mathbf{z}$, supplying α_f, α_r directly spares the network a relation the model already states. The objective

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_s \mathcal{L}_{\text{steady}} + \lambda_m \mathcal{L}_{\text{sym}} + \lambda_g \mathcal{L}_{\text{smooth}}, \quad (9)$$

with $\lambda_s = 0.1$, $\lambda_m = 0.05$, $\lambda_g = 10^{-4}$, adds three penalties to the one-step MSE [42], [21], each answering a weakness of training on one lap of ordinary driving. $\mathcal{L}_{\text{steady}}$ keeps the two residual channels jointly consistent with (1)–(2) on a quasi-steady mask, so they cannot imply an axle force

pair no tire could produce — and it is that force pair, not the state prediction, that stage 3 fits. $\mathcal{L}_{\text{sym}}$ enforces the odd symmetry $e(-\mathbf{z}) = -e(\mathbf{z})$ of a laterally symmetric vehicle, supplying from geometry the half of the input domain a circuit under-covers. $\mathcal{L}_{\text{smooth}}$ is a soft Lipschitz bound on how fast the residual may grow inside the range the extrapolation bound of Section V-C allows. Derivations, masks and the precomputation that makes the nominal-model half of (9) free are in the supplementary material.

One property of (9) is not a hyperparameter. $\mathcal{L}_{\text{data}}$ is a mean square over a one-step increment in m/s and rad/s, whereas the two balance residuals of (1)–(2) are a force and a moment; in those units the terms are four to six orders of magnitude apart and λ_s is not a relative weight at all. Measured at initialisation on a 30 s buffer of this vehicle, the unnormalised steady term contributes 99.9996 % of $\mathcal{L}$ against 0.0004 % for the data term, a ratio of $2.3e5$: such a residual is fitted to cancel the nominal model's own force imbalance and never to the data it was given. Each residual is therefore expressed as the state increment it implies, the unit the network's output is in,

$$r_{\text{lat}} = \frac{T_s}{m} \left(m \dot{v}_y + m v_x \omega - (F_{y,r} + F_{y,f} \cos \delta) \right), \quad (10)$$

and r_{yaw} likewise with T_s/I_z , before $\mathcal{L}_{\text{steady}} = \text{mean}(r_{\text{lat}}^2) + \text{mean}(r_{\text{yaw}}^2)$ is formed. The shares then become 98.53 % data and 1.47 % steady, recovering the meaning (9) gives λ_s . This is a correction to the objective as published, not an implementation detail.

2) *Temporal residual*: The s4 architecture replaces the memoryless MLP with an S4D diagonal state-space residual [23] over a sliding window of $W = 20$ samples (0.7 s at $T_s = 0.035$ s), so the residual can carry tire-relaxation and yaw-damping dynamics a per-sample map cannot. We claim neither the architecture, which is also in concurrent work [22], nor the $8.7\times$ reduction in its training cost, an exact algebraic rewrite of the same forward pass that changes

no identified coefficient; both are derived and measured in the supplementary material. Its accuracy effect is positive but measured offline (Section VI-K), and the identifiability result of Section V-B holds for either architecture.

B. Excitation-aware virtual rollout

This is the central correction. Consider stage 2 of Section III-D: a constant-speed rollout at v with the steering ramped to δ_{sweep} . In quasi-steady cornering the lateral acceleration a single-track vehicle develops kinematically is $a_y = v^2 \kappa$ with $\kappa \approx \delta/L$, so the largest friction coefficient the rollout can demand of the tire is

$$\mu_{\text{reach}} \approx \frac{v^2 \delta_{\text{sweep}}}{g L} \quad (11)$$

Equation (11) contains no tire parameter: it is a property of the rollout configuration alone, and an *upper* bound rather than an equality, because $\kappa \approx \delta/L$ is the Ackermann curvature while the steady-state curvature a compliant vehicle holds, $\kappa = \delta/(L + K_{us}v^2/g)$, is smaller. Reading it as a ceiling is all the argument needs: if even the ceiling sits below D , the peak is certainly not reached, the synthetic data never leaves the linear ramp of (4), only the product BCD is identifiable (Section III-C), and the bounded optimiser resolves the residual freedom on a box edge — the signature measured in Section VI-D.

The baseline implementation selects the rollout speed as $v = \text{clip}(\bar{v}_x, 2, 4) \text{ m/s}$, representative of a 1:10 platform. On our vehicle, with $\delta_{\text{sweep}} = 0.4 \text{ rad}$ and $L = 1.5334 \text{ m}$, (11) gives $\mu_{\text{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than 0.43 g whatever the tire is. At 1:10 scale, with $L \approx 0.32 \text{ m}$, the same speed yields $\mu_{\text{reach}} \approx 2.0$ and the pathology is invisible.

1) *Correction:* The rollout speed is now taken from the speed the vehicle was actually driven at, clamped by configuration bounds (lower bound 15 m/s, no upper bound), and the pipeline warns whenever (11) evaluates below the prior's D , turning a silent degeneracy into a startup diagnostic. The same bound sets a data-collection speed: identification data should be gathered at $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$.

C. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure: the rollout queries the residual network at steering angles it has never seen, the observed $|\delta|$ on a pure-pursuit lap staying below $\approx 0.06 \text{ rad}$ while the sweep ran to 0.4 rad . The fit reads the network's extrapolation as additional grip, and because each iteration's answer became both the next iteration's nominal model *and* its regularisation target, the error compounds: D walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, a claimed 1.7 g on a vehicle measuring 1.0 g in steady state.

Two changes close the loop:

- 1) **Extrapolation bound.** The sweep terminates at $\delta_{\text{sweep}} = \text{clip}(\lambda_e \cdot P_{99}(|\delta_{\text{obs}}|), \delta_{\min}, \delta_{\max})$ with $\lambda_e = 1.5$, $\delta_{\min} = 0.05 \text{ rad}$, $\delta_{\max} = 0.54 \text{ rad}$, so the network is never queried more than 50% beyond the

steering range it was trained on. Both $\delta_{\max}$ and the inherited $\delta_{\text{sweep}} = 0.4 \text{ rad}$ lie above the plant's 16.0° (0.279 rad) mechanical lock, by a factor of 1.9 and by 43% respectively. That is deliberate: the sweep is a *virtual* rollout, and clamping it to the lock would cap μ_{reach} a second time, by the actuator instead of the rollout speed. In practice the binding limit is the data-driven $\lambda_e P_{99}(|\delta_{\text{obs}}|)$ term. The residual is nonetheless evaluated outside the plant's reachable steering set, the extrapolation recorded in Section A, A6.

- 2) **Frozen prior.** The Tikhonov target is frozen at the configured prior Φ_p^0 for all iterations, while the optimiser's *start point* continues to track the previous iteration. The per-axle objective is

$$\min_{\phi} \sum_k \rho(r_k(\phi)) + \sum_j w_j (\phi_j - \phi_j^0)^2, \quad (12)$$

with r_k the force residual, ρ a soft- ℓ_1 robust loss and w_j scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The regulariser exists to hold the coefficients the manoeuvre does not excite, E above all, at their nominal value instead of on a box edge.

D. What the rollout fit identifies, and why the loop must be damped

Two further properties belong to the loop itself, and together they account for everything the corrected pipeline does with D and with the linear range.

1) *The rollout fit sees only the product BCD :* In the virtual-data step the low-slip degeneracy of Section III-C is stronger, and structural rather than excitation-dependent. Stage 3 does not observe a force; it reconstructs one from the rollout's own quasi-steady yaw balance (7), which on the synthetic trajectory reduces to

$$\frac{F_{y,f}}{F_{z,f}} = \frac{v_x \omega}{g \cos \delta}, \quad \frac{F_{y,r}}{F_{z,r}} = \frac{v_x \omega}{g}, \quad (13)$$

so the only quantity (12) ever compares against is the lateral acceleration the rollout realises — produced by the *previous* coefficient set through its slope $F_z BCD$. The fit's residual therefore depends on (B, C, D) only through their product, and D is the free coordinate of the hyperbola $BCD = \text{const}$: not separable from B , and stopping wherever the box constraint the other factors reach first puts it. Replayed on five identification buffers of a live $\mu = 1.05$ run (Section VI-K), the product is recovered tightly and repeatably, $BCD F_{z,f}$ over 18.9–21.1 kN/rad against a plant measured at 20.4 kN/rad from a prior 35% below it, while the D reaching that same product ranges over 0.69–2.00.

D must therefore not be asked of this fit at all. It is measured on the *recorded* trace by the brush estimator of Section V-F3, which fits $F_y(\alpha)$ directly and is not subject to (13), then held fixed while B , C and E carry the stiffness the rollout does determine (Algorithm 1, lines 3–5 and 14).

2) *The loop is not a contraction:* Stage 4 makes each iteration's answer the nominal model stage 2 rolls out, and stage 3 then fits the trajectory that model produced, over a steering range already extrapolating $\lambda_e = 1.5 \times$ past the observed one. The fit reads its own forces back, so the loop is a positive feedback path on its own extrapolation error rather than a contraction, and iterating it is not a refinement: replayed offline on five recorded buffers, axle force RMSE against the plant's telemetry degrades monotonically with iteration count, 38.9, 53.0, 58.5 and 70.4 N at the front for $N_{it} \in \{1, 2, 3, 6\}$. Pinning D does not remove this; it relocates it into B , C and E .

The remedy is to damp the step, not to shorten the loop. Stage 4 becomes a relaxed fixed-point iteration,

$$\Phi_p^{(i)} \leftarrow (1 - \beta) \Phi_p^{(i-1)} + \beta \tilde{\Phi}_p^{(i)}, \quad (14)$$

with $\tilde{\Phi}_p^{(i)}$ the solution of (12) at iteration i and $\beta = 0.20$; $\beta = 1$ recovers the inherited update exactly. The sweep over β (Section VI-K) carries the control the contribution needs: $\beta = 0$ freezes the coefficients at the pinned prior, performing no identification at all, and is the worst setting. The iterations do carry information; they have to be damped for it to survive.

E. Pacejka fit

Fitting applies the steady-state force recovery of (7) to the synthetic data under box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle, over B , C and E with D held at the value Section V-F3 put there. Relative to the baseline we (i) raised the jittered multi-starts from 1 to 8 and (ii) added a derivative-free simplex [43] and a global differential-evolution solver [44] alongside trust-region reflective [45], all through the same optimisation library [46]; population-based global search on Magic-Formula coefficients follows Cabrera *et al.* [47]. The prior carries the measured tire of Table II. One caveat on (i): `num_starts` is honoured by the trust-region and simplex solvers only, differential evolution being a population search that ignores it, so the multi-start claim describes the Baseline-NN-MSE arm and not the shipped Ours arm. Neither change addresses the degeneracy, and Section VI-B verifies both as secondary: at a 4 m/s rollout every tire tested fits back as $\hat{D} \approx 0.5$ with coefficients railed, and the answer moves neither with the multi-start seed nor with a jitter seven times wider.

F. Measurement-side corrections

Three corrections apply to the measurement chain rather than to the estimator.

1) *Achieved versus commanded steering:* α_f in (3) was built from the `/drive` setpoint. The measured achieved/commanded ratio is 0.76, so α_f was systematically $\approx 25\%$ too large and the fitted front stiffness $\approx 20\%$ too small (13725 N/rad against 17198 N/rad fitted to the simulator's own per-wheel forces over the same run); using `feedback/steering_angle` recovers 18486 N/rad. All three are effective slopes over that run's load range and so fall below the $\bar{F}_z = 1$ value of Table II; the comparison is between them.

2) *Mass and geometry:* m and l_{wb} are taken from the measured static wheel loads (Table I) rather than from the configuration file, and $l_{wb} \equiv l_f + l_r$ is enforced by a regression test, because (7) uses the wheelbase for the normal loads and the axle split for the force division, and the two disagreeing is a silent bias.

3) *Proprioceptive friction warm-start:* The inherited pipeline had no prior for the peak factor: D started from a configuration constant and was left to the fit. What follows recovers one from the sensors the vehicle already carries, in four steps.

a) *Axle forces and slip angles without a force sensor:* Newton–Euler on the single-track body of (1)–(2), solved for the axle forces rather than the accelerations, gives

$$F_{y,f} = \frac{I_z \dot{\omega} + l_r m a_y}{L \cos \delta}, \quad F_{y,r} = \frac{l_f m a_y - I_z \dot{\omega}}{L}, \quad (15)$$

which needs the IMU's lateral acceleration and the yaw acceleration and no tire model. Vertical loads carry the longitudinal transfer $F_{z,f/r} = m(g l_{r/f} \mp a_x h_{cg})/L$ [34]. Yaw acceleration is the derivative of a local polynomial fit to the yaw rate (Savitzky–Golay, window 21, order 2) [48] rather than a difference, because $I_z \dot{\omega}$ is a $\approx 10\%$ term in $F_{y,f}$ here and a differenced odometry yaw rate puts its noise straight into it. Slip angles come from (3) on the buffer already collected. Without an IMU the accelerations are differentiated from the states by the same filter; with one, the RMS difference between the two is logged as a bias diagnostic.

b) *Joint fit of stiffness and peak friction:* Per axle, cornering stiffness and peak friction are fitted jointly to the Fiala brush model [25], [49], [4]

$$F_y = \mu F_z (1 - (1 - \xi)^3) \text{sgn } \alpha, \quad \xi = \min\left(\frac{C_\alpha |\tan \alpha|}{3\mu F_z}, 1\right), \quad (16)$$

by bounded nonlinear least squares with a soft- ℓ_1 loss [46], [45]. The brush model replaces (4) here because it has two parameters, both of which the manoeuvre can support, against the four of the model whose non-identifiability at low slip this paper is about. A two-stage fit is avoided because estimating C_α first, on data already slightly nonlinear, biases it.

What beats the utilisation bound is that μ enters (16) through the *curvature* of $F_y(\alpha)$ and not only its asymptote. Writing $r = |F_y|/(C_\alpha |\tan \alpha|)$ for the force's departure from its own linear extrapolation, (16) inverts to

$$\mu = \frac{2k}{3 - \sqrt{3(4r - 1)}}, \quad k = \frac{C_\alpha |\tan \alpha|}{3F_z}, \quad (17)$$

so a 30% departure from linearity ($r = 0.7$) already pins μ , well below the limit. Equation (17) is used only for the excitation diagnostics.

c) *When the estimate is released:* Equation (17) states its own failure mode: $d\mu/dr$ grows without bound as $r \rightarrow 1$, i.e. as the curve stays linear. The estimate is released only when four conditions hold: absolute grip demand $\max |F_y|/F_z \geq 0.40$, peak friction utilisation $\max |F_y|/(\hat{\mu} F_z) \geq 0.40$, relative precision $\sigma_{\hat{\mu}}/\hat{\mu} \leq 0.15$ from the Gauss–Newton covariance, and $\hat{\mu}$ not resting on its own search bound; otherwise the

module logs which condition failed and falls back to the utilisation bound.

The first two are the same quantity with and without the fit in the denominator, and both are needed. The utilisation test divides by the *fitted* $\hat{\mu}$ of the same least-squares call, so it is circular: an axle whose μ comes out too low reports a correspondingly high utilisation and clears the gate on its own error — on a live run a buffer realising only 0.31 of F_z returned $\hat{\mu} = 0.539$ at utilisation 0.58 and passed, against a plant of 1.05. The absolute ratio has no such freedom, and on the five buffers of that run it separates usable from unusable exactly: $\{0.146, 0.314\}$ rejected, $\{0.541\text{--}0.677\}$ accepted. The precision and railing conditions remove what survives both. The 40% floor is not tuned here; it is the level the lateral-dynamics friction-estimation literature converges on, as surveyed by Acosta *et al.* [16], who report estimators failing below $|a_y| = 0.4g$ “due to the proximity of the tyre force curves in the low slip regions” and a 40–80% utilisation working range. Samples with $|a_x| > 0.35g$ are dropped rather than corrected, since the axle longitudinal force split a friction-ellipse correction needs is not observable from this sensor set; samples below 2.5 m/s are dropped because (3) is ill-conditioned there; and samples whose force opposes their own slip angle are dropped as sign errors or noise.

d) *Applied two-sided, then pinned:* The released $\hat{\mu}$ replaces D on both axles, in either direction, and D is then held fixed for the whole co-identification loop. Four points fix that rule.

First, both axles take the *front* fit. The rear slip angle does not carry δ , so on this trajectory the rear force curve has too little curvature to support (17) and its μ comes out high: 1.495/1.383/1.257 rear against 1.011/0.964/0.837 front on the three admitted buffers of a $\mu = 1.05$ run, with $\sigma_{\hat{\mu}}/\hat{\mu} = 0.105$ at the rear, inside the gate. This is not a yaw-inertia error: in quasi-steady cornering $F_{y,f}/F_{y,r} \rightarrow l_r/l_f$, the same ratio as $F_{z,f}/F_{z,r}$, so $\mu_f = \mu_r$ whatever I_z is, and sweeping I_z over 30–80 kg m² only widens the axle gap from 0.368 to 0.550. There is one road surface, so the axle the manoeuvre identifies speaks for both.

Second, it is two-sided rather than a floor: the floor rule it replaces was protecting against a D that *moved* between cycles, not one that was low, and pinning supplies that directly without a one-way bias a low-friction surface makes untenable. Third, it is pinned and not merely seeded, since (13) leaves D free along $BCD = \text{const}$ and on a live run a 0.785 seed ended at the 2.000 bound within one cycle. Fourth, an axle with no released $\hat{\mu}$ holds the D it came in with rather than reverting to fitting it, since Section V-D1 shows there is nothing in the rollout to fit it against.

Two downstream mechanisms make the fixed D what the interface needs. The frozen Tikhonov target of (12) is taken from the warm-started prior, so whatever lands there is both initial guess and regularisation target for all N_{it} iterations; and the controller re-ingests the reference speed profile at $\min(a_{y,\max}^{\text{cfg}}, a_{y,\max}^{\text{max},\eta})$ on every accepted update, so a D moving *within* a cycle rewrites the reference under the running controller. The requirement is that D not move during a cycle, not that it only ever rise.

Pinning puts the whole weight of the result on the gate, and σ_{μ}/μ from the fit covariance is a *precision* figure, not an accuracy one: a wrong mass, wrong yaw inertia, IMU bias or road bank all fit tightly and wrongly through (15), axle force scaling linearly with m . Here the configuration file’s 240.0 kg against the 269.6 kg the wheels actually carry biased μ 11% low until the measured value replaced it (Table I); likewise the configured wheel coefficient is not the effective one the physics step uses, the server multiplying it by the road surface’s factor of 0.70, so an estimate validated against the former would be judged 30% short while being correct. Of the inputs to (15) only I_z now remains unmeasured; it enters $F_{y,f}$ at the 10% level (Section A).

G. Deployment

The identifier runs as a single ROS 2 node whose acquisition and state-machine timer ticks at 50 Hz (the 20 Hz quoted elsewhere is the controller’s period), cycling a rolling FIFO buffer through four states: collect, train, await the controller’s acknowledgement, run, re-identifying every $T_{\text{reid}} = 30$ s from the newest buffer. Three interface details matter for reproduction. The achieved road-wheel angle is used where published, falling back to the commanded setpoint on staleness with the substitution logged. The friction warm start runs on *every* cycle: restricting it to the first, as an earlier version did, does not hold D still, since the node re-reads the static `pacejka_model` prior each cycle, so every later cycle cold-starts D from a value unrelated to the road — on a live run cycles 2–6 then walked D to its 2.0 bound. And the coefficients are submitted over a service call whose non-acknowledgement leaves the previous model in place, so a rejected identification degrades to the last accepted model rather than to none; the peak-friction outputs are published on a latched diagnostic topic so a late-starting consumer sees the current value.

That acknowledgement also arms the scoring: the node’s own online metrics and the parameter forward to the passive benchmark node both fire from the acknowledgement handler, not from the end of training. A submission the controller *rejects* on its admissibility gate (Section VII) is a model the car never drove on, and the seconds between a fit completing and its acknowledgement landing are seconds still driven on the previous model; scoring either attributes to one model the behaviour of another.

H. Validation protocol

Three levels of validation are used, in increasing strength:

- 1) **One-step prediction** on a held-out run, as in the baseline: RMSE and R^2 of v_y and ω under $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. A dedicated ROS 2 node extends this to configurable multi-step horizons (1/5/10 steps) with Euler/Heun/RK4 integration and online statistics in Welford’s form [50]. Samples are admitted only above 1 m/s, the gate the identification data uses. This is not a division-by-zero guard: since $\alpha = \arctan(v_y/v_x)$, a few tenths of a metre per second of lateral drift saturates the tire curve and the model predicts yaw rates of several

rad/s against a ground truth of a few tenths, so a looser gate lets the low-speed bootstrap dominate a metric meant for the racing regime.

- 2) **Direct force comparison** against `feedback/tire_forces`: the identified curve is evaluated at the measured (α, F_z) and compared per axle against the measured F_y , and against the plant's own curve (8) at the same points (Table II), which separates the error of the fit from the error of representing a PhysX tire.
- 3) **Closed-loop tracking** under the MPC using the identified model: mean and maximum lateral deviation, its time-weighted integral, heading error and solver cost, split by active controller so the pure-pursuit acquisition phase is not scored against a model it does not use. Lap counting runs from the first crossing of the raceline's $s = 0$, so the drive from the spawn point is an untimed out-lap.

All three levels score the *accepted* model only, for the reason given in Section V-G.

VI. EXPERIMENTS AND RESULTS

A. Setup

All experiments run on the architecture of Section IV: the Formula AI vehicle in CARLA [8] driven through ROS 2 Humble, the server in synchronous mode at a 0.033 s fixed physics step. The identifier collects 30 s of data at 50 Hz and re-identifies from the newest buffer, matching the baseline's data budget [6]. Ground truth for forces, slip angles and normal loads is the simulator's own per-wheel telemetry, which the identifier never reads.

1) *Surface*: Every run uses a *static* surface: the per-wheel coefficient is held at its configured 1.0, and the road-surface factor of 0.70 PhysX applies on top of it makes the effective coefficient 0.70, constant to the last digit the simulator reports over all 10744 and 12025 force-validation samples of the two runs. It is deliberately *not* the $\mu = 1.05$ surface the tire prior was measured on (Section VI-A4), so a pipeline that cannot move off its prior has nowhere to hide. It remains a *single* surface: time-varying friction is out of scope, and the adaptivity claim a friction sweep would support is not made (Section VIII-C).

2) *Reference trajectory*: Both runs follow the same optimised raceline (Fig. 3), produced with the TUM global race trajectory optimisation package [51] under its minimum-curvature formulation [33] on a recorded centreline of the CARLA silverstone circuit: a closed 5828.2 m lap of 2916 waypoints at 2.0 m spacing, reference speed 9.62–25.78 m/s, minimum corner radius 18.7 m ($|\kappa|_{\max} = 0.0536/\text{m}$). Its peak lateral demand is $\max(v_x^2|\kappa|) = 4.98/\text{m/s}^2$ (0.51 g), which saturates the flat 5.0 m/s² ggV envelope the optimiser was given and stays at roughly 73 % of the 0.70 g this surface sustains. It replaces the earlier trajectory, whose 1.22 g demand the admissibility gate of Section VII rejected against the same plant; that regeneration, not a relaxed gate, unblocked the closed-loop runs.

TABLE III
THE TWO CONFIGURATIONS.

Setting	Baseline-NN-MSE	Ours
Residual architecture	MLP (baseline)	S4D (s4)
Residual objective	One-step MSE	Physics-informed, (9)
Pacejka solver	Trust-region [45]	Differential evolution [44]
Friction warm start	Off	On; D pinned per axle
Update relaxation β	1.00 (undamped)	0.20

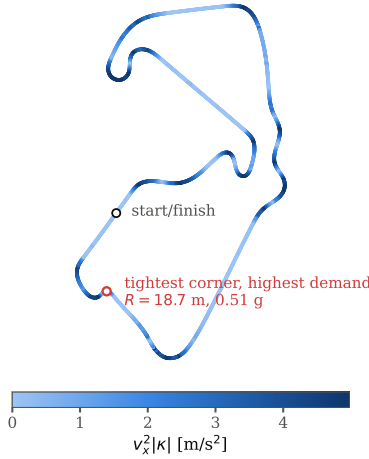
Everything not listed is identical between the two, β is the relaxation of (14); 1.00 is the inherited undamped update.

3) *The two configurations*: Two configurations of the same pipeline are run back to back on that trajectory, three timed laps each, with an untimed out-lap from the spawn point to the raceline's $s = 0$. They differ in the five settings of Table III and in nothing else; vehicle, map, trajectory, controller, data budget, loop count and box constraints are shared. *Baseline-NN-MSE* is the inherited configuration (MLP residual, one-step MSE objective, local trust-region fit, no friction warm start, undamped update) evaluated at full scale on the corrected rollout, so the comparison isolates those five settings rather than reproducing the published pipeline in full (Section VI-I); *Ours* is the shipped configuration. Since the five settings move together, what follows is a configuration comparison, not a per-factor ablation, which Section VI-K supplies offline. Nor is it a test of this paper's central contribution: the excitation-aware rollout is active in *both* arms, so the closed-loop comparison measures the five secondary settings on top of it and the correction itself is evidenced in Section VI-B.

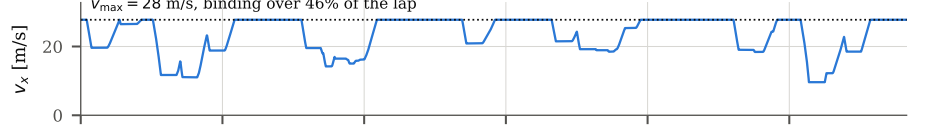
4) *Provenance of the tire prior*: The prior Φ_p^0 both arms start from, and that the Tikhonov term of (12) pulls toward, is the measured tire of Table II, obtained by fitting (4) to CARLA's own per-wheel telemetry over a dedicated limit-excitation run that reached $\alpha = 0.37$ rad, axle $\mu = 1.05$ and 1.50 g, by differential evolution, at 135 N front and 85 N rear residual RMSE. Three consequences follow. The prior is not independent of the validation data, since that is the same channel the force and friction results are scored against. The manoeuvre that produced it is one the pipeline is not allowed to run: the raceline peaks at 0.51 g and never approaches the tire's peak, the excitation deficit this paper is about. And, decisively, the prior was measured on a surface these runs do not use: it carries $D = 1.009/1.002$ and axle slopes of 13.2 kN/rad front and 13.9 kN/rad rear against the 0.70 and the 17.4 and 16.6 kN/rad this surface realises (Section VI-D). Returning it unchanged is a 44 % grip overstatement and a 16–24 % stiffness deficit, visible in every metric of Table V; to score well an arm has to *move*, in the right direction, from a prior that is wrong.

5) *What is measured*: Three families of quantity are recorded against the plant's own telemetry: the identified tire model (coefficients, axle force, peak friction), the one-step state prediction, and closed-loop MPC tracking. Every figure is exported with the CSV holding its samples, and the cross-run comparison reads those CSVs rather than re-deriving any metric, so a figure and the table cannot disagree.

(a) raceline, 5.83 km lap, coloured by lateral demand



(b) reference speed profile



(c) lateral demand against available grip

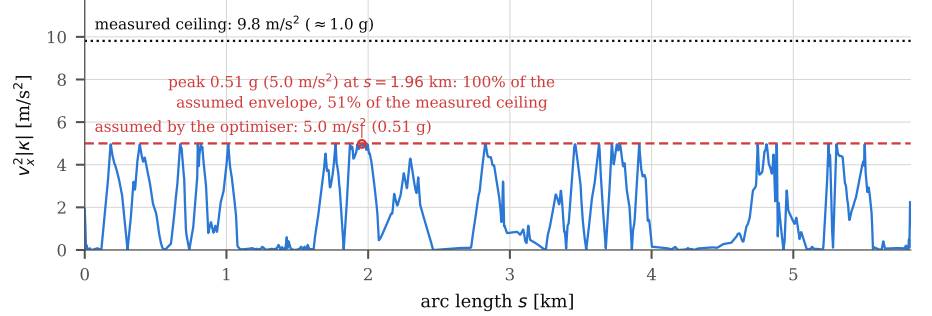


Fig. 3. The optimised raceline used for both runs, generated with the TUM global race trajectory optimisation package [51] under its minimum-curvature formulation [33] from a recorded centreline of the CARLA silverstone circuit. (a) Track layout, coloured by lateral demand, with the marked corner ($R = 18.7$ m) the tightest on the lap. (b) Reference speed profile, 9.62–25.78 m/s. (c) Lateral demand $v_x^2 |\kappa|$ along the lap against the flat 5.0 m/s^2 envelope handed to the optimiser (dashed) and the grip the vehicle has on this surface (dotted, ≈ 0.70 g, measured). The profile saturates the envelope and no segment exceeds the plant's grip.

B. Identifiability of the peak factor against rollout excitation

The claim of Section V-B concerns the rollout configuration alone, so it is tested where the answer is known exactly rather than in CARLA. A tire with a known peak factor D is rolled out through the synthetic sweep of stage 2 (the shipped rollout, residual network set to zero) and fitted back with the shipped solver, *unregularised* and started from the true coefficients, so nothing it loses is lost to the prior, the start point or a local minimum; everything it loses is lost to excitation.

Three axes are swept, rollout speed v , sweep amplitude δ_{sweep} and true D , each cell repeated over five noise realisations. The configuration axes are swept one at a time, because μ_{reach} is a function of both and a claim that it, and not the speed, governs the fit cannot be tested by moving the speed alone. The repetition is over noise rather than over the solver because the noise-free fit is exactly seed-invariant here: five multi-start seeds return the same coefficients to four decimal places, and so does a jitter width seven times wider. The injected noise is Gaussian on the rolled-out states at a one-step persistence RMSE measured on CARLA runs of this vehicle, $\sigma_{v_y} = 0.009 \text{ m/s}$ and $\sigma_{\omega} = 0.005 \text{ rad/s}$, and is what the intervals in Table IV are computed over; the front and rear fits share one rollout, so the interval is taken over the five realisations, not the ten fits.

The grid says three things. First, the inherited 4 m/s rollout is uninformative about D at *every* grip level tested: the 0.70, 1.05, 1.40 and 1.80 tires all come back as $\hat{D}$ between 0.50 and 0.59, with two to seven of the eight coefficients on a bound, so the fit reports μ_{reach} and not the tire. Second, the boundary is $\mu_{\text{reach}} \gtrsim D$ and not a speed: at 6 m/s , where $\mu_{\text{reach}} = 0.96$, the 0.70 and 1.05 tires are recovered to within 3 % while the 1.40 and 1.80 tires are 13 % and 30 % short. The boundary is soft: the 1.05 tire is recovered to 1 % at $\mu_{\text{reach}} = 0.96$, slightly *below* its own D , because a rollout stopping just short of the peak still traverses the curvature separating

B , C and D . The bound predicts where the information runs out, not a step. Above $\mu_{\text{reach}} = 1.7$ every tire is recovered to within 4 % with no systematic railing. Third, the failure is not the solver's: by the seed invariance above, multi-start, a global solver or a better start point cannot recover a coefficient the data does not contain, which is why Section V-E reports both as secondary. The one caveat is at the top of the sweep, where a rollout far past the peak over-reports a low-grip tire ($\hat{D} = 0.92$ for a 0.70 tire at 20 m/s), the Magic Formula's falling branch being fitted to data the plant's own curve does not produce; the extrapolation bound of Section V-C keeps the shipped configuration out of that regime.

Panel (b) sweeps the other argument of (11), δ_{sweep} from 0.03 rad to the shipped $\delta_{\text{max}} = 0.54 \text{ rad}$ at fixed $v = 15 \text{ m/s}$. Starving the amplitude destroys D just as starving the speed does, so speed is not the governing variable, and the claim is corrected in two ways. First, the failure is not a common bias: an under-speeded rollout returns *too little* grip (every tire back at $\hat{D} \approx 0.5$), an under-swept one too much, the 1.05 tire coming back as 1.30 and the 1.80 tire railing on the 2.0 bound at $\delta_{\text{sweep}} = 0.06 \text{ rad}$, with two of eight coefficients on a box edge in every cell. What the axes share is that $\hat{D}$ stops being a function of the tire, the direction of the error being set by the rollout: the signature of an unidentified parameter, not a biased one. Second, the axes do *not* collapse onto a common μ_{reach} , for the upper-bound reason of Section V-B: (11) uses the Ackermann curvature and so tracks the realised demand only while understeer is small, the realised $\max |v_x \omega|/g$ saturating near the tire's peak while μ_{reach} grows to 10.64 against a realised 1.40–2.18 at $v = 20 \text{ m/s}$. Read against the realised demand, both axes behave alike: recovery sets in where the rollout actually reaches the peak. Equation (11) should therefore be used as the pipeline uses it, a cheap screening bound conclusive when it falls below D , not a measure of excitation achieved.

TABLE IV
PEAK FACTOR $\hat{D}$ RECOVERED FROM A KNOWN TIRE, ALONG BOTH ARGUMENTS OF μ_{reach} .

swept	μ_{reach}	$\hat{D}$ (railed of 8) at true D of			
value		0.70	1.05	1.40	1.80
<i>(a) rollout speed v [m/s] swept at $\delta_{\text{sweep}} = 0.40$ rad</i>					
4	0.43	0.52±0.036 (1.8)	0.57±0.013 (2.8)	0.59±0.018 (3.8)	0.50±0.001 (7)
6	0.96	0.68±0.001 (1)	1.04±0.014 (1.2)	1.22±0.066 (1.8)	1.26±0.097 (2)
8	1.70	0.69±0.001 (1)	1.02±0.001 (0.8)	1.36±0.002 (0)	1.79±0.025 (1.2)
10	2.66	0.70±0.000 (0)	1.03±0.001 (0.4)	1.37±0.002 (1)	1.75±0.002 (0)
13	4.49	0.72±0.001 (0)	1.05±0.002 (0)	1.38±0.001 (0.4)	1.77±0.002 (0.4)
15	5.98	0.76±0.009 (1)	1.05±0.002 (0)	1.39±0.002 (0)	1.78±0.001 (0)
20	10.64	0.92±0.004 (1)	1.14±0.005 (0.4)	1.43±0.002 (0)	1.80±0.005 (0)
<i>(b) δ_{sweep} [rad] swept at $v = 15$ m/s</i>					
0.03	0.45	0.80±0.019 (2)	1.10±0.041 (2)	1.08±0.182 (2)	1.09±0.397 (2)
0.06	0.90	0.92±0.003 (2)	1.30±0.008 (2)	1.68±0.017 (2)	1.80±0.232 (3.4)
0.10	1.50	0.96±0.004 (2)	1.34±0.016 (2)	1.60±0.017 (1)	1.97±0.030 (1.4)
0.15	2.24	0.99±0.004 (2)	1.38±0.003 (2)	1.71±0.019 (2)	1.98±0.009 (1)
0.25	3.74	0.77±0.002 (2)	1.12±0.004 (2)	1.81±0.020 (2)	2.00±0.000 (2)
0.40	5.98	0.76±0.009 (1)	1.05±0.002 (0)	1.39±0.002 (0)	1.78±0.001 (0)
0.54	8.08	0.83±0.004 (1)	1.08±0.002 (0)	1.38±0.001 (0)	1.78±0.008 (0)

Unregularised, started from the true coefficients. Each cell is $\hat{D} \pm$ the half-width of its 95% t -interval over the five noise realisations, with the mean number of the eight coefficients resting on a box constraint in parentheses. The two axes of one rollout are not independent, so each realisation contributes the mean of its own ($\hat{D}_f, \hat{D}_r$) and $n = 5$; per-axle means and intervals are in the supplementary material.

μ_{reach} is (11) at $L = 1.5334$ m; the inherited configuration is the first row of (a). It is the Ackermann upper bound, and above $\mu_{\text{reach}} \approx 2$ it detaches from what the rollout realises: at $v = 20$ m/s it reads 10.64 while the rollout's realised $\max |v_x \omega|/g$ is 1.40–2.18 across the four tires. A large μ_{reach} means “far past saturation”, not a demand the tire ever sees.

One property bounds how all of this should be read: the residual network is set to zero, so a Magic Formula is fitted to Magic-Formula data under exactly matched structure. The recovery is a best case, and the real pipeline's degradation at a given excitation is at least this bad.

C. Cross-run summary

Table V is the complete comparison. Ours is better on every quantity scored against the plant (peak friction, axle force on both axles, both one-step states, lateral tracking error in every phase, and both error tails) and behind only on RMS QP solve time, by 5% of a figure two orders of magnitude inside the control period. The margins are 9.0–12.1× on peak friction and axle force and 1.5× on MPC tracking error. The rest of this section reads those rows against the figures; Section VI-K attributes them.

D. Identified tire model

Figure 4 puts both identified curves against the plant's own curve (8) and lists the coefficients. Three things are visible.

First, no coefficient of Ours rests on a box constraint in any of its six accepted identifications; its last accepted set is $B = 12.890/13.746$ inside $[4, 20]$, $C = 1.240/1.248$ inside $[1.2, 2.2]$, $D = 0.768$ on both axles inside $[0.4, 2.0]$ and $E = -2.517/-1.836$ inside $[-3, 1]$. Baseline-NN-MSE puts D on its upper bound of 2.000 on both axles in every accepted identification and E_f on its lower bound of -3.0 likewise, its first identification additionally railing B on the lower bound of 4.0 on both axles. A peak factor of 2.000 against a plant realising 0.70 is less a 186% parameter error than the box-edge signature of Section III-C: the coefficient is not identified, and no per-coefficient bounds check distinguishes that from a

fit. Two coefficients of Ours need care. C settles 0.04 above its lower bound, inside the box but close enough to report as shaped by the constraint, consistent with its role in the BCD product of Section V-D1. E is not identified in either configuration: it shapes only the region around and past the peak, which neither the lap nor the bounded rollout reaches, and it is exactly what the frozen Tikhonov target of (12) holds near nominal — Ours ends at -2.52 against a -2.00 prior, where the unregularised arm ends on the bound.

Second, the linear range is identified rather than inherited. Every slope here is (5) for an identified set; for the plant it is the slope fitted to the simulator's own per-wheel forces over $|\alpha| \in [0.005, 0.02]$ rad, all at the static axle loads $F_{z,f} = 1373$ N, $F_{z,r} = 1274$ N, so the three are comparable. The plant realises 17.4 kN/rad front and 16.6 kN/rad rear over this run's load range, below the $\bar{F}_z = 1$ closed-form values of Table II for the reason its footnote gives. The prior carries 13.2 and 13.9, 24% and 16% low; Ours returns 16.9 and 16.8, 3.2% low front and 1.4% high rear, removing 87% and 91% of the prior's error. The product $BCDF_z$, the only quantity the rollout fit determines (Section V-D1), is what moves and then holds: the first cycle takes the front value from 13.2 kN/rad to 14.6, and the five cycles after the friction gate releases sit within 16.0–16.9 kN/rad, a spread of 5% about a plant at 17.4. Baseline-NN-MSE returns 37.9 and 38.4, 118% and 132% high, which is $D = 2.0$ multiplying a B the fit chose against it.

Third, both sets keep the understeer sign ($l_r C_r > l_f C_f$) and pass the stability gate of Section VII-A, at margins of 7.3% for Ours ($l_f C_f = 12.45$ against $l_r C_r = 13.36$ kN m/rad) and 9.1% for Baseline-NN-MSE (27.98 against 30.52), against 2.5% for the plant. What separates them is the grip ceiling: (6) gives 2.00 g for Baseline-NN-MSE against 0.77 g for

TABLE V
CROSS-RUN COMPARISON OF THE TWO CONFIGURATIONS.

Metric	Ours	Baseline-NN-MSE
<i>Closed-loop tracking (3 timed laps)</i>		
Timed laps	3	3
Run duration [s]	580.4	554.6
Best lap [s], wall clock	142.84	135.82
Mean lap [s], wall clock	154.47	145.39
RMS e_y , overall [m]	0.256	0.319
RMS e_y , MPC [m]	0.151	0.224
Max $ e_y $, MPC [m]	0.683	0.776
ITAE e_y [m s ²]	16041	25502
IAE e_y [m s]	60.2	87.2
IAE/ T [m]	0.104	0.157
ITAE/ $(T^2/2)$ [m]	0.095	0.166
RMS heading error [rad]	0.0206	0.0215
Max heading error [rad]	0.315	0.310
Emergency halts	0	0
MPC solve time, RMS [ms]	0.399	0.380
MPC solve time, max [ms]	2.604	2.568
<i>Axle lateral force against per-wheel telemetry</i>		
Front RMSE [N]	27.2	329.6
Rear RMSE [N]	33.2	348.4
Front R^2	0.993	-0.160
Rear R^2	0.987	-0.461
Front max abs. error [N]	264	1295
Rear max abs. error [N]	135	1316
<i>Peak friction against the plant's $\mu = 0.70$</i>		
Front μ RMSE, whole run	0.145	1.300
Rear μ RMSE, whole run	0.142	1.300
Front / rear μ RMSE, after first release	0.078	1.300
<i>One-step state prediction</i>		
v_y RMSE [m/s]	0.014	0.156
ω RMSE [rad/s]	0.049	0.104
v_y persistence RMSE [m/s]	0.011	0.017
ω persistence RMSE [rad/s]	0.005	0.007
Skill vs. persistence, v_y	-0.27	-8.18
Skill vs. persistence, ω	-8.8	-13.9

Ours, a plant sustaining 0.71 g and a trajectory demanding 0.51 g . The baseline's ceiling is a claim the plant cannot honour by a factor of 2.8 and the prior's 1.01 g one it cannot honour by 1.4; only the identified set is admissible on the quantity the controller consumes.

All three are readable because the lap now traverses part of the curvature: over the force-validation samples $P_{99}(|\alpha|) = 0.047 \text{ rad}$ (2.7°) and $\max |\alpha| = 0.102 \text{ rad}$ (5.8°) against the 8.0° at which the plant's curve peaks at $\mu = 0.70$, and axle utilisation $\max |F_y|/F_z$ reaches 0.71 front and 0.72 rear, against 0.49 for the same trajectory on the $\mu = 1.05$ surface of the earlier runs. This is still not a limit manoeuvre — the rollout reaches up the curve and the warm start carries the independent information about μ — but it suffices for the friction gate to release, which on the $\mu = 1.05$ runs it did not.

E. Peak friction

Table V reports the error of the identified peak factor against the effective friction the plant reports per wheel, and Fig. 5 its evolution over the run. This is where the pinned- D design of Section V-F3 is directly observable, because the arm starts 44 % away from the answer.

Ours publishes six accepted identifications. The first carries $D_f = 1.009$, $D_r = 1.002$: the friction gate rejected that buffer, so the axles held the prior they came in with, the hold-rather-than-fit rule of Section V-F3d and not a fitted value. From

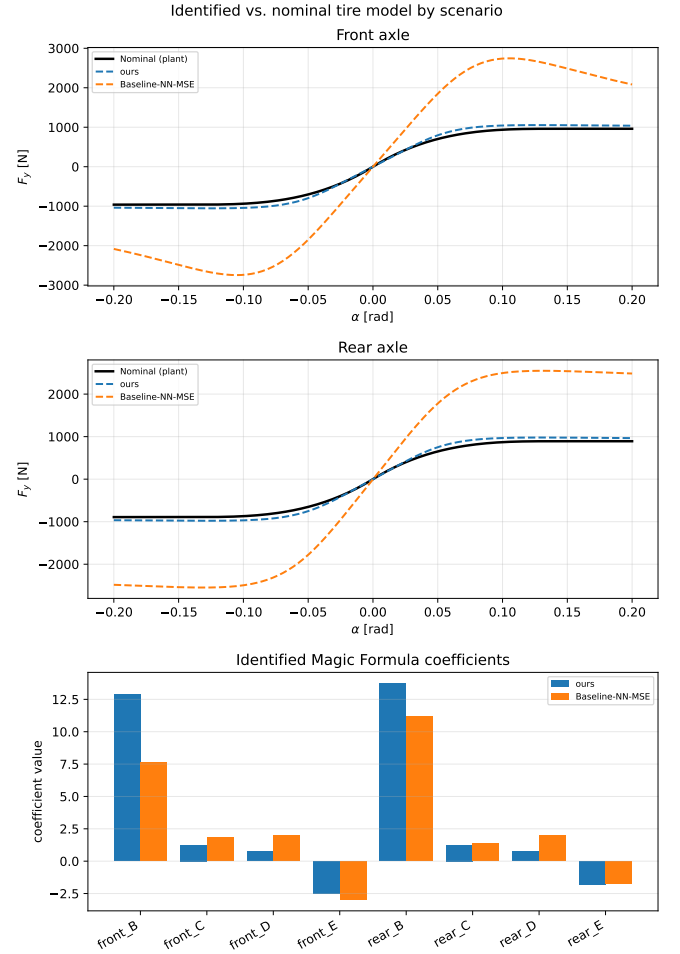


Fig. 4. Identified tire models against the plant. Top and middle: the plant's own curve (8) at the measured static load of each axle (black) against each run's last accepted identification, over $\alpha \in [-0.2, 0.2]$ rad. Bottom: the identified Magic Formula coefficients. Baseline-NN-MSE's accepted peak factor sits on its box constraint $D = 2.0$ on both axles in every cycle, so its curve carries roughly 2.8 times the plant's force at the edge of the plotted range, while Ours tracks both the plant's initial slope and its peak. Note that the per-wheel lateral force and the per-axle vertical load exported in the underlying CSV are not on the same basis; the axle forces plotted here and every force metric in Table V are axle sums.

$t = 140.6 \text{ s}$ the gate releases on every subsequent cycle and the published peak factor drops to 0.779, 0.821, 0.749, 0.756 and 0.768, front and rear alike, against a plant at 0.700: 10.7 % high on average with a spread of 0.072, from a prior that was 44 % high. The whole-run μ RMSE of Table V is dominated by the first 140 s at the prior; over the 440 s after the first release it is 0.078 on both axles. Baseline-NN-MSE returns $D = 2.000$ on both axles in every accepted identification and never leaves the bound, a constant μ error of 1.300, so Ours is $9.0\times$ better on the whole-run figure and $16.7\times$ better once its warm start has released — a difference worth 1.23 g of claimed grip against a plant that sustains 0.71.

Two qualifications belong on the Ours column. The peak factor is *measured*, by the brush estimator on the recorded trace, then held, not recovered by the Pacejka fit, which Section V-D1 shows cannot recover it. And the residual 10.7 % bias is one-signed, consistent with the brush model's known

tendency to over-read μ where the curve is not fully traversed, which at $\max |F_y|/F_z = 0.71$ it is not.

F. Axle lateral force

The identified curve is scored at the measured (α, F_z) against the plant's own lateral force, summed per axle (Table V; Fig. 6 for the error distribution, bar charts and time series in the supplementary material). *Ours* leads on both axles by an order of magnitude in RMSE and in the error tails. The baseline's negative R^2 means its curve predicts the plant's axle force worse than the mean of that force does, the force-metric reading of a set whose initial slope is more than twice the plant's.

This is the linear-range result of Section VI-D seen as force. At $P_{99}(|\alpha|) = 2.7^\circ$ the metric is dominated by cornering stiffness and says little about the peak, so the models separate almost entirely on slope — but because D multiplies that slope in (5), *Baseline-NN-MSE*'s railed D is the direct cause of its force error rather than an overstatement of grip the linear range hides, the opposite of what the earlier $\mu = 1.05$ runs suggested.

The comparison is also why the relaxed update of (14) belongs in the method rather than a configuration file: pinning D alone does not produce these numbers, since with $\beta = 1$ the same arm degrades monotonically with iteration count (Section VI-K).

G. One-step state prediction

Level 1 of the validation protocol is reported in Table V against a persistence baseline (next state equals last measured state) written by the same benchmark node. *Ours* leads on both signals, by an order of magnitude on lateral velocity (R^2 0.998 against 0.709) and by a factor of two on yaw rate (R^2 0.894 against 0.514). In Fig. 7 the separation on v_y is in the centre of the distribution, while on ω the two share a centre and differ in tail mass.

Neither model beats persistence at the 0.035 s model step: the skill scores $1 - \text{RMSE}/\text{RMSE}_{\text{pers}}$ are negative for both configurations and both signals. We report them rather than R^2 alone because a one-step metric on a smooth trajectory rewards not moving — over one 0.035 s step the plant's own states barely move. One-step accuracy is therefore weak evidence for either tire model, and the gap is read only as confirming the direction the axle-force and closed-loop comparisons establish. The state traces against ground truth are in the supplementary material.

H. Closed-loop tracking

Both runs finish three timed laps with zero emergency halts and the same finite-state sequence (bootstrap pure pursuit, running pure pursuit, handover, running MPC), one transition of each kind and no fallback after handover, read off each run's exported state timeline, which records the handover at $t = 79.1$ s for *Ours* and $t = 64.0$ s for *Baseline-NN-MSE* and no state change thereafter. (The aggregated transitions column of the cross-run summary is not populated and is omitted from Table V.) Under MPC, where the identified model

enters the loop, *Ours* tracks at 0.151 m RMS lateral error against 0.224 m, a 32 % reduction, with maximum 0.683 m against 0.776 m (Fig. 8). ITAE is $\int t|e_y| dt$, neither scale-free nor comparable across runs of different length, and these runs are 580.4 s and 554.6 s long, so Table V reports it beside two normalised forms from the same samples, IAE/ T and ITAE/ $(T^2/2)$. All three order the runs alike and at the same magnitude, so the normalisation does not carry the result. The ITAE split confirms that the whole difference sits in the MPC phase: the pure-pursuit phases, which do not use the identified model, give 110 and 107, within 3 % of each other and two orders of magnitude smaller. The signed error distribution (supplementary material) shows the mechanism as a narrower distribution about a comparable centre, sample standard deviation 0.256 m against 0.318 m and mean 0.021 m against 0.024 m.

The QP cost is unchanged: RMS and worst-case solve times are two orders of magnitude below the 50 ms control period and both slightly in the baseline's favour, by 5 % and 1 %. The identified coefficients change the model the QP carries, not its size.

Lap time is not read as a result: timing runs on the ROS wall clock while the simulator advances its fixed step faster than real time, at a ratio that differed between the runs, so the lap times in Table V are not simulated lap times and their 5–6 % spread is not a vehicle-performance difference.

I. Baseline pipeline versus this work

The supplementary material tabulates the pipeline as published [6] against the pipeline as it stands here, on every axis that differs. The residual model, the data budget, the four stages and the steady-state force recovery are inherited unchanged; what differs is the configuration of the virtual-data step, the relaxation on the loop closing it, the regularisation topology, the measurement chain and the contract at the output. Table III covers a subset of those axes, since the rollout correction is active in both runs.

J. Peak-friction warm start on a synthetic plant

The friction warm start is evaluated on a synthetic single-track plant with brush tires rather than in CARLA, because only a synthetic plant supplies a μ that is known exactly and an excitation level that can be swept independently of it. The plant is driven by a swept-sine steering input for 40 s at 50 Hz with true $\mu = 0.70$, the same effective peak as the CARLA surface, and $C_\alpha = 12.1F_z$; the steering amplitude sets how far up the tire curve the manoeuvre reaches (Table VI).

At 63 % utilisation, a level an ordinary lap reaches, the utilisation bound is 37 % low while the brush fit is within 0.2 %; across true $\mu \in \{0.7, 1.05, 1.4\}$ at matched utilisation it stays within 0.2 %, and C_α within 0.1 %, on noise-free data. With odometry and IMU noise injected the error is grip-dependent, the noise being absolute while the available force is not: at $\mu = 0.70$ it is 20 % at 63 % utilisation, 6 % at 80 % and 0.4 % at the peak. Below the absolute 0.40 floor on $|F_y|/F_z$ the gate suppresses the fit and reports the bound, so the failure is visible rather than silent — on a 0.70 surface that floor is

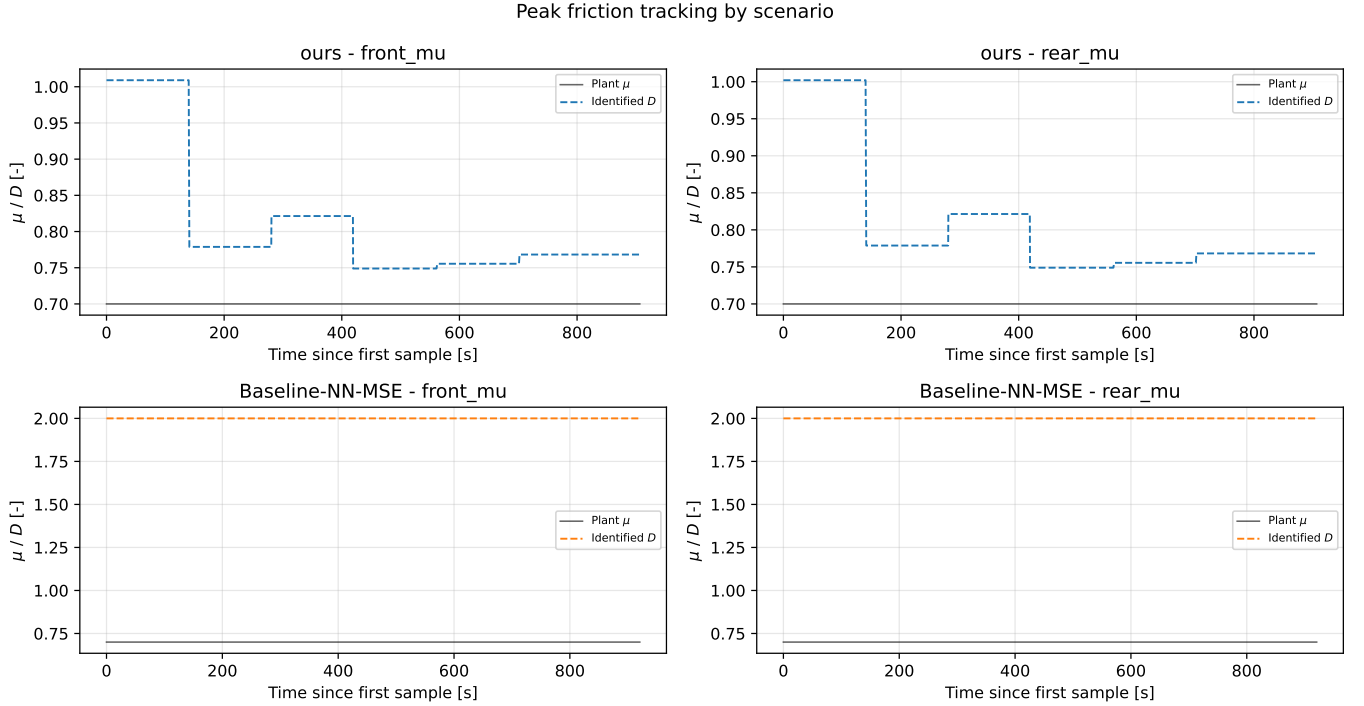


Fig. 5. Peak friction over the run, one row per configuration: the plant’s effective μ (black) against the identified peak factor D published at each accepted identification. The surface is static, so every departure from the black line is estimator error. Ours publishes the $\mu = 1.05$ prior while the friction gate is closed, then steps down onto the plant within 11 % at the first release and holds there for the remaining five cycles, while Baseline-NN-MSE sits on the $D \leq 2.0$ box constraint for the whole run.

TABLE VI
PEAK FRICTION RECOVERED FROM A SYNTHETIC BRUSH-TIRE PLANT
(TRUE $\mu = 0.70$), AGAINST EXCITATION.

δ amplitude	Grip utilisation	Utilisation	Brush fit (16)
0.033 rad	45 %	0.312	gated off
0.047 rad	63 %	0.442	0.699
0.060 rad	80 %	0.557	0.699
0.090 rad	100 %	0.700	0.697

The utilisation estimator tracks the grip the manoeuvre *used*, while the brush fit of (16) recovers the peak from the curvature of $F_y(\alpha)$ and is therefore flat in excitation.

The first row is refused by the absolute gate $\max |F_y|/F_z \geq 0.40$ of Section V-F3: at $\mu = 0.70$, 45 % utilisation is only 0.31 of F_z . The utilisation bound is reported instead.

reached at 57 % utilisation, not 40 %, which is why the first row of Table VI is refused. The gate behaves the same way at both ends on the CARLA run: it rejects the first buffer, and on the five it accepts returns μ within 11 % of the plant at an absolute axle utilisation of 0.54–0.68. That 11 % bias against the 0.2 % of the synthetic plant is the cost of a real rollout, a reconstructed $I_z \dot{\omega}$ term and a curve traversed only to 0.71 of F_z , and it dominates the identified peak factor’s error.

K. What carries the result

The two arms differ in five settings at once, so that comparison establishes an ordering and attributes nothing. This subsection attributes it, at the cost of leaving the loop: every number below is an *offline replay* of the identification stage on recorded buffers.

1) *Protocol*: `nn_train` is replayed on the five 30 s identification buffers each arm collected during its own run, and each configuration is scored with the quantity the closed-loop tire-force benchmark itself reports: the identified Magic Formula evaluated at the plant’s own measured slip angles, against its own measured lateral forces, summed per axle. No synthetic plant and no re-derived metric are involved, so these numbers and Table V’s are on the same scale. One caveat for anyone reproducing this from the exported curve-validation files: the lateral force they carry is *per wheel* while the vertical load beside it is the *axle* static load, and fitting them as written returns every cornering stiffness a factor of two low.

2) *Damping the co-identification loop*: Table VII sweeps β of (14) at six iterations throughout; two columns are the argument. At $\beta = 1.00$, the inherited undamped update, axle force RMSE is 70.4 and 62.3 N, reached by degrading monotonically from 38.9 and 26.3 N at a single iteration: the loop is not refining the estimate, it is amplifying its own extrapolation. At $\beta = 0.00$ the coefficients never move from the pinned prior — no identification at all — and that is the *worst* column, 92.5 and 71.8 N. The minimum sits between them at $\beta = 0.20$ ($23.5 \text{ N} \pm 1.5 \text{ N}$ over seeds 0, 3 and 7), and the identified BCD rises monotonically with β across the sweep, so the damping trades how much of each fit is accepted against how much extrapolation error comes with it rather than simply interpolating toward the prior. Six iterations are therefore defensible on evidence, but only damped.

3) *Pinning the peak factor*: Replayed on five buffers of a $\mu = 1.05$ run, letting the fit move D gives a mean $|D - \mu|$

Tire lateral force error distribution by scenario

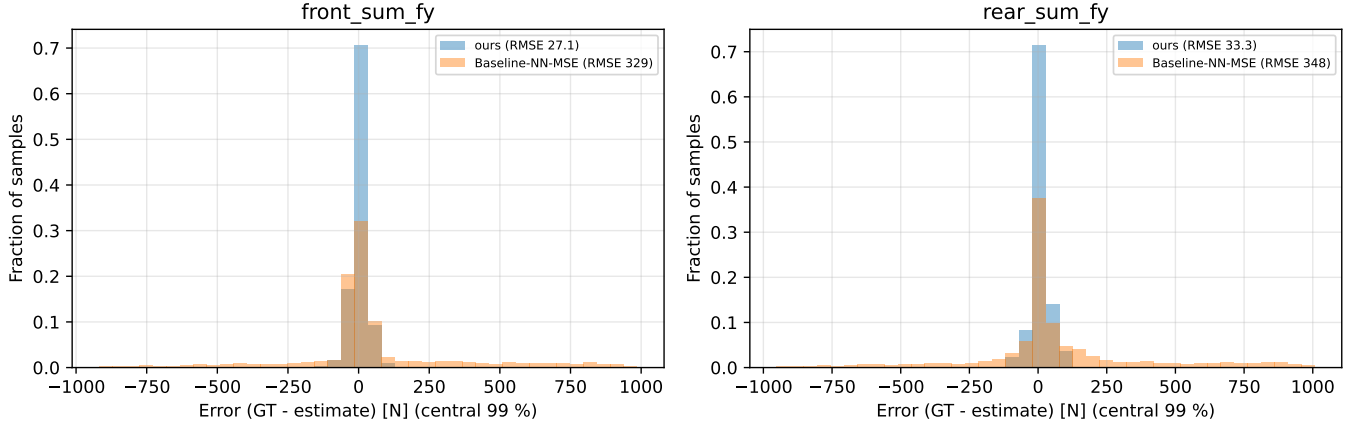


Fig. 6. Distribution of the axle lateral-force error (truth minus estimate), shared bin edges over the pooled central 99% and normalised to the sample count of each run. Both are centred, and *Baseline-NN-MSE*'s distribution is roughly an order of magnitude wider at every quantile (maximum absolute error 1295 N against 264 N front, 1316 N against 135 N rear).

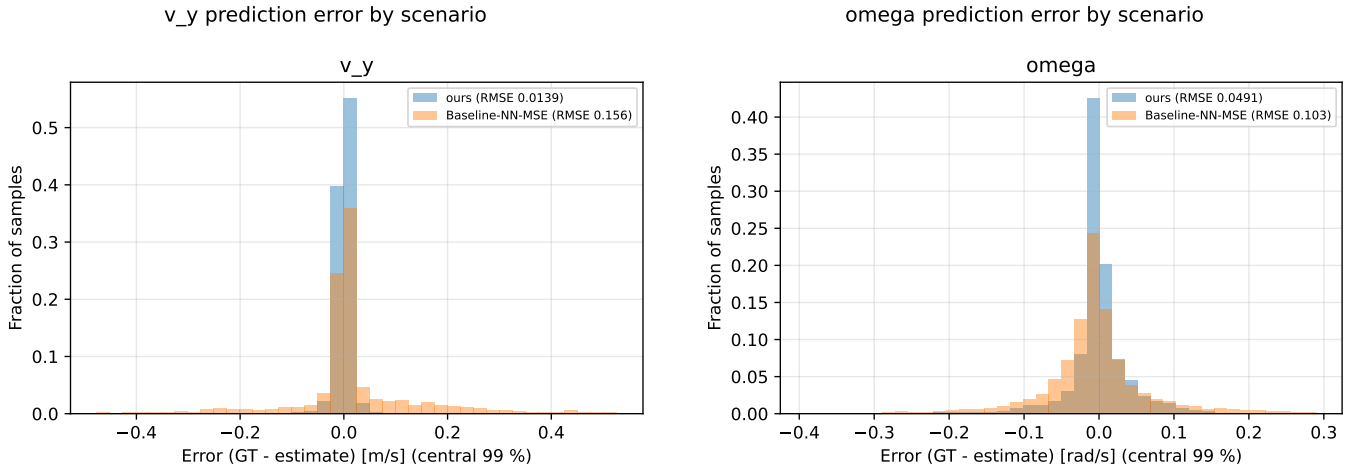


Fig. 7. One-step prediction error (truth minus estimate). Left: lateral velocity, where *Ours* is an order of magnitude tighter (0.014 m/s against 0.156 m/s RMSE). Right: yaw rate, where the difference between the configurations is tail mass rather than centre. Neither model beats its own persistence baseline at the 0.035 s model step.

TABLE VII
RELAXATION SWEEP. FIVE RECORDED BUFFERS, SIX ITERATIONS,
OFFLINE REPLAY.

β	0.00	0.05	0.10	0.15	0.20	0.25	0.30	0.50	1.00
Front F_y RMSE [N]	92.5	68.3	48.6	33.9	28.0	31.2	36.4	55.6	70.4
Rear F_y RMSE [N]	71.8	49.0	31.3	19.8	19.4	26.0	30.2	38.4	62.3
Front BCD [rad]	7.82	8.82	9.66	10.36	10.88	11.32	11.65	12.75	12.42

$\beta = 1.00$ is the inherited undamped update and $\beta = 0.00$ the pinned prior, i.e. no identification. 0.20 ships.

of 0.585 and a maximum of 0.950, with the identified front stiffness spread over 32.9 kN/rad across the five. Pinning only the axles whose brush fit was released reduces those to 0.191, 0.378 and 32.3; additionally holding the axles with no released measurement, the shipped rule, reduces them to 0.084, 0.213 and 18.5 kN/rad. The middle row is the informative one: releasing an axle helps only where the gate fires, and what removes most of the remaining error is refusing to let an

unmeasured D be fitted at all. The same replay is the source of the BCD degeneracy measurement quoted in Section V-D1.

4) *Normalising the physics-informed objective*: The unnormalised form of (9) puts 99.9996% of the objective on the dynamics-residual term at initialisation against 0.0004% on the data; with the T_s/m and T_s/I_z scaling of (10) the shares become 1.47% and 98.53%. The correction is necessary but not sufficient alone: applied without the damping it makes the force metric slightly worse (77.1 and 76.2 N against 70.4 and 62.3 N), because an objective that now fits the data also supplies a sharper trajectory for the undamped loop to amplify. With the damping in place the two arrive at 28.0 and 19.4 N and $R^2 = 0.986$ at the front, against 92.5 and 71.8 N and 0.870 for the unidentified prior.

5) *Two settings that do less than claimed*: Two of the five settings of Table III deserve a weaker statement than the closed-loop comparison alone would support. The S4D residual is worth 21% on the force metric at matched settings and

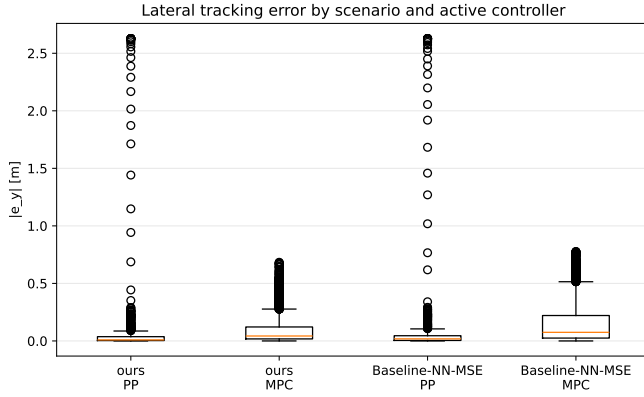


Fig. 8. Absolute lateral tracking error by configuration and active controller. The pure-pursuit boxes cover the acquisition phase before handover and do not use the identified model. The MPC boxes do.

a single iteration, 35.3 N against the baseline MLP’s 44.7 N as a mean over the two axes — the ablation the earlier version of this work listed as unrun, positive in sign, but an offline replay and not a closed-loop result. The residual *objective* is a wash: with the normalisation and the damping both in place, *physics_informed* and *plain mse* are indistinguishable at 23.7 N each. We keep the physics-informed objective because it costs nothing and is the paper’s stated formulation, and do not claim it buys accuracy on this manoeuvre, because it measurably does not.

L. Identification cost

One S4D identification, six co-identification iterations on a 30 s lap, takes 20.9 s end to end, inside the 30 s re-identification period. The reduction from the 181.4 s it originally took is exact: no epoch count, iteration count, early-stopping rule or loss weight changed, and the identified coefficients agree to four decimal places. The per-phase breakdown, the device split, and the caveat that the stage micro-benchmarks and the end-to-end totals are separate measurements rather than a decomposition of each other, are in the supplementary material.

VII. INTEGRATION WITH THE CONTROLLER

The identified model leaves the pipeline through one interface. A nonlinear MPC path-tracking controller (acados [31], partial-condensing HPIPM [32], horizon $N = 20$ at a 20 Hz control period) carries the same single-track model as the identifier and consumes Φ_p directly: the eight coefficients enter the prediction model’s axle forces through (4), pushed over a ROS 2 service at the end of each identification cycle, and nothing else about the controller changes. We claim no contribution to the controller design; it appears here as the consumer that defines what an admissible identification is.

A. Admissibility gates

A coefficient set can lie strictly inside every box constraint and still be unusable. We observed three distinct ways, each requiring a gate on a *derived* quantity rather than a coefficient:

- 1) **Grip ceiling versus trajectory demand.** The controller computes $\max_k (v_{x,k}^2 |\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (6) falls below it. On the earlier trajectory the degenerate fit reported a 0.40 g ceiling against a 1.22 g demand and produced 41.89 m of cross-track error at 27 m/s, against 0.03 m for a physically plausible set ($D = 1.5$) on the same trajectory, at the same speed and through the same controller; the same identified set gives 0.16 m at 11.1 m/s and 0.43 m at 15 m/s, the v^2 scaling the Introduction describes. All these figures come from an offline closed-loop harness (20 s on the reference, handover at $t = 5$ s) rather than a controlled pair of simulator runs, and are quoted as the observation that motivated the gate, not as a result; they are not confounded with the transport-delay correction below, which was already in place when they were measured. The gate is one-sided by construction and catches only a ceiling that is too low, so Baseline-NN-MSE’s 2.00 g (Section VI-D) passes it.
- 2) **Understeer sign and critical speed.** A set with $l_f C_f > l_r C_r$ whose v_{crit} falls inside the controller’s speed range makes the prediction model open-loop unstable, and the resulting $\rho(A_d) > 1$ is raised to ρ^N in the condensed QP. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\text{crit}} = 14.5$ m/s at the measured geometry and mass of Table I, inside the controller’s speed range on every trajectory in this paper.
- 3) **Integration stiffness.** Even a stable set can be stiff relative to the control period: a single explicit RK4 step at $dt = 0.05$ s gave $\rho(A_d) = 547$ at 5 m/s for one identified set. The model sub-steps adaptively, sizing the sub-step from its own fastest lateral mode $|\lambda| \approx (C_f + C_r)/(mv_x) + (l_f^2 C_f + l_r^2 C_r)/(I_z v_x)$ so that $|\lambda|h \leq 1.5$, restoring $\rho \approx 1.000$ across the speed range over the shipped $N = 20$ horizon. The cost is carried by the solve: over the reported runs QP setup and solve took a median of 0.36 ms and a 95th percentile of 0.62 ms against the 50 ms control period.

The gates are the contract between identification and control: an identification that passes is a model the controller can be held responsible for, one that fails is reported as unidentified rather than silently deployed. Two further controller-side changes were required before any identified model could be evaluated at full-scale speeds — reference speeds are clamped at ingest, and the solved state is rolled forward by the measured transport delay before each solve, which took cross-track error at 27 m/s from 130.55 m to 0.38 m — recorded for completeness, not as identification results.

VIII. DISCUSSION AND LIMITATIONS

A. What generalises

The reachable-friction bound (11) is kinematic: it involves no tire parameter, no simulator-specific quantity and no property of the residual model, so it applies to any implementation of the virtual-steady-state-data strategy of [6], in simulation or on hardware. Its practical content is a design rule — *the*

rollout must be configured to demand at least the friction one intends to identify — and where it cannot, D should be reported as unidentified rather than fitted. The diagnostic signature, coefficients resting exactly on box constraints, is likewise generic and should be checked and logged by any bounded tire-fitting routine, being otherwise indistinguishable in the output from a successful fit.

The two structural properties of Section V-D generalise on the same grounds. That stage 3's residual depends on B , C and D only through their product follows from the quasi-steady force recovery (7) defining the virtual-data step, not from any tire or simulator, so any implementation inherits it and with it the conclusion that the peak factor must come from the recorded trace. That the loop is not a contraction follows from stage 4 alone, and the remedy is one line reducing to the inherited behaviour at $\beta = 1$. What does not generalise is the value 0.20, a property of this vehicle's extrapolation error; what does is that $\beta = 1$ is not defensible and $\beta = 0$ worse still.

The insufficiency of per-coefficient box constraints for deployment admissibility is also platform-independent: both failures we encountered, a set that satisfies every box constraint while degenerate and one that satisfies every box constraint while pairwise oversteering and unstable inside the controller's operating range, are properties of the parameterisation rather than of CARLA. Deep Dynamics [21] constrains coefficients inside the network; we argue a second, derived-quantity check belongs at the interface where the model is handed to a controller.

B. What does not generalise

Three results are properties of this plant and must not be read as statements about tires. First, the numerical tire values: the plant is a PhysX wheeled-vehicle model with a configured friction coefficient, so Table II describes *that* plant, and agreement with it shows the identifier recovers the plant it observes, not anything about real tire behaviour. Sagmeister *et al.* [28] show sensitivity to model fidelity is largest near the acceleration limits, where this study operates, and R-CARLA [27] exists because CARLA's native dynamics are a known weak point for racing-grade work; hardware replication is therefore necessary before any claim about identification accuracy on a real vehicle. Second, the gap between configured wheel friction and realised peak axle friction (1.0 against 0.70 on the runs of Section VI, 1.5 against 1.00–1.05 on the earlier ones) is an artefact of the server multiplying the configured coefficient by the road surface's own, a measured constant 0.70 on this map; we report it because an identifier tuned to converge on the configuration file's number will converge on the wrong one. Third, the 0.76 achieved/commanded steering ratio arises from CARLA's Ackermann controller and this vehicle's steering configuration; the lesson, identify against the achieved actuator state, is general, the magnitude is not.

C. Limitations

- 1) **One run per configuration, so no metric in Table V carries an uncertainty.** Every value is a point estimate

from a single 580.4 s and a single 554.6 s run. The 0.151 m against 0.224 m RMS tracking difference in particular has no confidence interval and no evidence that it exceeds run-to-run variation, which is not measured here; the order-of-magnitude force and friction margins are large enough that such variation is unlikely to reverse them, but that is an argument, not a measurement. Nothing prevents repetition — the sweep is one command and the scenario file already carries the per-arm parameter sets — so this is a gap in the evidence rather than the method.

- 2) **Single surface, vehicle and trajectory.** That the surface differs from the one the prior was measured on (Section VI-A4) makes the identification a real result rather than a restatement of the prior, but it is *not* a demonstration of adaptivity: the identifier is never asked to follow a *change* in friction. The runtime friction override and the step and decay schedules the scenario file already carries support that experiment, arguably the strongest argument for on-track identification, and we do not perform it here.
- 3) **The friction estimator's excitation sweep is synthetic** (Table VI), chosen because μ is known there and excitation can be swept independently of it. On the CARLA runs the warm start is exercised at the one excitation level the lap supplies, returning μ 11 % high against the 0.2 % of the synthetic plant, so the excitation dependence of that bias is not mapped against the simulator's per-wheel forces. The published peak factor is exactly this estimate, so the bias passes through undamped.
- 4) **The rear axle is not identified on this trajectory** and the front-axle μ is applied to both (Section V-F3d). The quasi-steady argument making $\mu_f = \mu_r$ holds on a homogeneous surface and would not survive a split- μ one, which this sensor set cannot detect in the first place.
- 5) **The two configurations differ in five settings at once** (Table III), so Section VI-C attributes nothing to the residual architecture, the loss, the solver, the warm start or the relaxation individually. Section VI-K does attribute them, but by offline replay on recorded buffers, so its ordering is not a closed-loop ordering — the status, too, of the S4D figure there (21 % on force RMSE at matched settings), which we report as a measured ablation and not as a reproduction of the closed-loop S4 improvement of [22]. The excitation-aware rollout, this paper's central contribution, is active in *both* arms, so the closed-loop comparison is not evidence for the identifiability correction; the evidence for that is the sweep of Section VI-B.
- 6) **The physics-informed objective buys nothing measurable here.** Once its units are reconciled with the data term and the loop is damped, it is indistinguishable from a plain one-step MSE on this manoeuvre (Section VI-K). It is retained because it is the paper's stated formulation and costs nothing, not because it is shown to help.
- 7) **Longitudinal dynamics are out of scope.** As in the baseline, only lateral dynamics are identified. Combined

slip and load transfer are neglected, and the simulator's longitudinal wheel force channel is drivetrain torque rather than a contact-patch force, so it is unusable as ground truth for a longitudinal extension without further work.

IX. CONCLUSION AND FUTURE WORK

This paper studied the identifiability of a learning-based on-track Pacejka identification pipeline [6] at full vehicle scale, using a CARLA simulation architecture that exposes the plant's own per-wheel forces as ground truth. Transplanted unchanged from a 1:10 platform, the pipeline fails in a specific, diagnosable and correctable way: a loss of excitation created by its own virtual-data step, which no change of solver, bounds or tuning removes, and whose reachable friction $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}} / (gL)$ depends on the rollout configuration and not the tire. A rollout speed appropriate at 1:10 caps μ_{reach} at 0.43 at full scale, below the plant's peak, so the peak factor is not identifiable from the synthetic data and the bounded optimiser resolves the ambiguity on a box edge. Two further properties of the loop follow from the same analysis. The fit's residual depends on B , C and D only through their product, because stage 3 reconstructs its forces from the rollout's own quasi-steady yaw balance and so observes nothing but the lateral acceleration the previous coefficient set produced; the peak factor is therefore not recoverable from the rollout at any excitation and must be measured on the recorded trace and held. And because each fit is handed the trajectory its own previous answer generated, the loop is a positive feedback path rather than a contraction, so stage 4 is reformulated as a relaxed fixed-point iteration.

The correction comes with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved rather than the commanded road-wheel angle, measured mass and geometry, a friction warm start identified from the curvature of the axle force curve and pinned for the loop, a relaxation $\beta = 0.20$ on the coefficient update, and a dynamics residual normalised to the units of the data term it is added to. Together these produce identifications with no railed coefficient and the correct understeer sign. Run against the inherited configuration over three timed laps on a 0.70 surface, deliberately not the 1.05 the configured prior was measured on so that returning the prior scores badly, the corrected pipeline recovers axle cornering stiffness to 3.2 % and 1.4 % and the peak factor to 11 %, against an inherited configuration that rails D on its 2.0 box constraint in every cycle and overstates stiffness by 118–132 %; axle force RMSE is over ten times lower and MPC lateral tracking error 32 % lower, at unchanged solver cost. Those numbers are an identification and not a preserved prior, with three qualifications: the peak factor is identified by the brush estimator on the recorded trace, not by the Pacejka fit, which cannot identify it (Section V-D1); one surface is not adaptivity, which needs a friction *change* the runs do not contain; and each configuration is a single run, so the closed-loop margins are observations rather than estimates with a confidence interval. That the peak factor is recoverable from the rollout *when the excitation reaches it* is shown on

a known tire in Section VI-B, not on these laps. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities at that interface instead; the earlier reference trajectory, which demanded 1.22 g from a vehicle sustaining 1.0 g , was regenerated because that gate refused it rather than loosened for it.

A. Future work

a) Immediate:

- 1) Run the ablations of Section VI-K in the loop. The relaxation sweep, the D -pinning comparison and the S4D and objective ablations are offline replays, enough to attribute the result but not to confirm the ordering survives closed-loop coupling. Each is a scenario-file row.
- 2) Sweep the friction rather than merely mismatching it. These runs put the prior 44 % away from a *static* surface, showing the pipeline moves on evidence but not that it tracks; the step and decay schedules already in the scenario file are the intended vehicle, and the strongest available argument for identifying on track at all.
- 3) Excite the peak. The lap reaches $P_{99}(|\alpha|) = 2.7^\circ$ and 5.8° at most, against a plant whose curve peaks at 8.0° on this surface, at an axle utilisation of 0.71. That is enough for the friction gate to release, and it is also where the brush estimator's 11 % bias comes from; a trajectory or injected manoeuvre traversing the peak would remove it.
- 4) Run the noise sweep of [6] on this platform using the bridge's odometry noise model, with a nonlinear least-squares identification of the same data as the comparison arm.
- 5) Cross-check the friction warm start against the simulator's per-wheel forces across a pace sweep, including the σ_μ/μ gate boundary and the IMU-versus-kinematics cross-check guarding the mass and inertia assumptions. Since the published peak factor now *is* that estimate, its bias is the pipeline's bias.
- 6) Report lap time on the simulator's own clock, which currently makes lap time unusable as a comparison metric (Section VI-H).
- 7) Measure I_z rather than assuming it (Section A).

b) Beyond:

- 1) **Excitation-aware data collection.** Equation (11) inverts into a requirement on the manoeuvre. A controller that deliberately inserts bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would move the pipeline from passive on-track identification toward active, safety-bounded experiment design: optimal input design [52] under a safety constraint, where the literature for this step already exists.
- 2) **Uncertainty-aware output.** The gates of Section VII-A are binary. Reporting a posterior over the coefficients, through the Bayesian variational path already implemented alongside the deterministic solvers, would let

the controller weight the model by its confidence rather than accept or reject it.

- 3) **Hardware replication.** The decisive test of every claim in Section VIII is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, and that is what makes the simulation study a necessary precursor rather than a substitute.

APPENDIX A

ASSUMPTIONS AND REPRODUCIBILITY NOTES

The following are assumptions or design choices rather than measured facts, listed so that a reader can judge how far each result depends on them, and so that each can be discharged by a specific measurement.

A1. Yaw inertia I_z

$I_z = 51.1 \text{ kg m}^2$ is carried in the model file and is *not* measured. It enters (2) directly, so it scales the yaw-rate residual the network is trained on, and it enters the stiffness-adaptive sub-stepping of Section VII-A through $(l_f^2 C_f + l_r^2 C_r)/I_z$. *How to discharge:* apply a known yaw moment (a step steer at constant speed, axle forces read from the force feedback topic) and fit I_z from the measured $\dot{\omega}$; or read the value the physics engine uses from the actor's inertia tensor.

A2. Centre-of-gravity split

$l_f = 0.738142 \text{ m}$, $l_r = 0.795362 \text{ m}$ are taken from the controller configuration and confirmed by the vehicle owner. The measured static load split (51.2/48.8 front/rear) implies a marginally different split than the geometry does (51.9/48.1), a 1.4% disagreement, unresolved and below the level at which any conclusion here would change.

A3. CG height and load transfer

$h_{cg} = 0.3 \text{ m}$ is carried in the model file and unused: the single-track model neglects load transfer, following the baseline, and normal loads are static. On a full-scale vehicle at 1g this is a stronger approximation than at 1:10, and a candidate explanation for part of the residual the network absorbs.

A4. Steering-angle sign and units

`feedback/steering_angle` is published in degrees and treated as the front road-wheel angle, positive left, matching the ROS body frame; it was verified to agree with the front-left wheel joint state to within 0.5%. Left and right road wheels are not distinguished, the single-track δ being this single reported value.

A5. Steady-state force recovery

Equation (7) is not a force measurement. Being fully determined by $v_x \omega$ it misattributes transients, and it is applied only to the *synthetic*, quasi-steady rollout, where the steady-state assumption is constructed rather than hoped for. Its adequacy was nonetheless checked on measured data against the simulator's own axle force (correlation 0.993, slope 0.94); we treat that as validating the relation, not a steady-state assumption on recorded laps.

A6. The residual network is not the plant

The corrected model $\hat{x}_{k+1} + e_k$ is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section V-C limits but does not eliminate this: the sweep still reaches $1.5\times$ the observed 99th percentile of $|\delta|$. The choice of 1.5 is empirical, sufficient to reach the tire's peak at the corrected rollout speed while removing the observed drift; it is not derived.

A7. Simulator determinism and timing

CARLA runs in synchronous mode with a 0.033 s physics step, but sim time advances faster than wall time (observed $\approx 1.9\times$ on our hardware). All identification data is stamped from the simulation clock.

A8. Software configuration of record

Identification and control workspace: ROS 2 Humble, Python 3.10 (required, since ROS 2 Humble ships CPython 3.10 ABI extensions only), PyTorch with CUDA on an RTX 2080-class GPU. The bridge is built and sourced as a separate overlay on that workspace, so its message packages (`sim_manager_msgs`) resolve for the ground-truth forces; CARLA server 0.9.16 / Unreal Engine 4.26; **acados 0.5.5** with partial-condensing HPIPM. The identifier's acquisition and state-machine timer runs at 50 Hz (20 Hz is the controller's period) with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at 5×10^{-4} , and the one-step model is discretised at $T_s = 0.035 \text{ s}$ (Section IV-E).

Random seeds. `nn_train` seeds PyTorch and NumPy at 42 for the single-member architectures used here (42+ i per ensemble member). `pacejka_solver.seed` is null in both shipped parameter sets, so the multi-start jitter, and for Ours the differential-evolution population, draw from an unseeded generator: the reported identifications are reproducible in distribution but not bit for bit. The sweep of Table IV is seeded explicitly and is bit-reproducible.

APPENDIX B

SIMULATOR-ROS 2 INTERFACE

Command path

The bridge runs in `ackermann_drive` mode, subscribing to a single `ackermann_msgs/AckermannDriveStamped` on `/drive` and forwarding speed and steering angle to CARLA's native Ackermann controller, whose server-side cascaded PID loops

TABLE VIII
ROS 2 INTERFACE USED BY THE IDENTIFICATION AND VALIDATION
PIPELINE. NAMESPACE IS /SIM UNLESS MARKED ABSOLUTE.

Topic	Type	Role
/odom	nav_msgs/Odometry	State v_x, v_y, ω ; twist is body-frame
feedback/ steering_angle	std_msgs/Float32 (deg)	Achieved road-wheel angle δ
feedback/imu	sensor_msgs/Imu	a_x, a_y for the friction warm-start
feedback/ tire_forces	sim_manager_msgs/ TireForces	Per-wheel $\alpha, F_z, F_y, \mu, \omega_w, \kappa$: ground truth
feedback/ vehicle_physics	std_msgs/String (JSON)	Live wheel parameters of (8)
feedback/speed	std_msgs/Float32	Forward speed echo
feedback/motors	std_msgs/String (JSON)	Per-wheel speed/torque/brake
/drive	ackermann_msgs/ AckermannDrive Stamped	Control command in
control/ tire_friction	std_msgs/Float32	Runtime grip override
/clock (abs.)	roscpp_msgs/ Clock	Simulation time base

are configured once at startup. No client-side steering PID is present, the commanded road-wheel angle passing through in radians, so the achieved/commanded ratio of Table I is a property of the vehicle and its steering curve rather than of a tuning choice in the bridge.

Feedback path

Table VIII lists the interface consumed by the identifier and its validation.

a) *Per-wheel ground truth:* feedback/tire_forces carries CARLA's own WheelTelemetryData: per-wheel lateral slip angle, normal load, lateral force, effective friction coefficient, wheel speed, longitudinal force and wheel torque, in wheel order [FL, FR, RL, RR]. No analytic tire model exists inside the bridge, so nothing on this topic can disagree with the physics the vehicle is driven by, which is what allows an on-track identifier to be scored against the forces the plant generated. Four measured properties bound what may be read off it. (i) Only lateral_force is force ground truth; it is the channel every validation in this paper uses. (ii) longitudinal_force is a friction-capacity report rather than a measurement, tracking μF_z and correlating with ma_x at 0.04, and wheel_torque is an exact identity of it, so neither carries independent information. (iii) Below 0.5 m/s the publisher zeroes slip angle and both forces while the normal load stays live, so a load-only gate admits standstill frames as perfect (0,0) samples; the identifier gates on the all-zero pattern as well. (iv) The slip angle needs no sign change across the boundary, verified rather than assumed.

Figure 9 states the conventions the rest of the paper assumes. The simulator and the identification stack do not share a handedness: CARLA inherits Unreal's left-handed world (X east, Y south, Z up, yaw positive clockwise), while every ROS 2 topic follows REP-103 [53], right-handed ENU for the world and forward-left-up for the body, and the identification equations are written in the body frame with α and F_y positive to the left, the SAE-style axis convention [54], [4]. The bridge

is a per-channel conversion rather than a single transform: position and yaw flip sign, twist is additionally rotated into the body frame by q^{-1} , the IMU's a_y flips, the steering *command* flips on the way out, and the tire telemetry's slip angle does *not*, because the frame flip and CARLA's own sign convention for lat_slip cancel. That last is the trap: a sign error there is invisible in the states and inverts the identified curve, so it was checked against a slip angle rebuilt from the chassis state in a steady 4 m/s turn at 0.20 rad, equal in magnitude to within 2 % front and 15 % rear.

Extensions made for identification

Six changes support this study and are listed for reproducibility: the existing but unsubscribed feedback/steering_angle is now consumed by the identifier (with configurable units, scale and staleness timeout, falling back to the /drive setpoint); runtime overrides control/tire_friction and control/drag_coefficient write straight to the live VehiclePhysicsControl, so surface friction can change within a run without respawning; the torque curve, zero-throttle damping and wheel tire_friction were tuned against a low-speed hold test (a wheel friction above ≈ 2.0 on a steered wheel scrubs hard enough at creep speed to stall the drivetrain intermittently), and these values define the plant being identified; telemetry runs on a dedicated 10 Hz thread off the control loop, since each tick costs several blocking CARLA RPCs; feedback/tire_forces was extended with the effective friction coefficient, the wheel speed and a kinematically recomputed slip ratio $\kappa = (\omega_w r_w - v)/v$, the last because CARLA's own long_slip field is inconsistent with the wheel speed and radius the same message reports (+0.177, i.e. hard braking, on a wheel turning 8.99 rad/s at a steady 2.33 m/s cruise, whose kinematic ratio is -0.034); and the live wheel parameters of (8) are published on feedback/vehicle_physics, so the reference curve of Table II cannot go stale when the vehicle configuration changes.

APPENDIX C

REFERENCE IMPLEMENTATION

The identification package, the ROS 2 bridge and the controller stack used in this paper are available as On-Track-SysID, extended from [6] (upstream: <https://github.com/ForzaETH/On-Track-SysID>), CARLA_ASU_BRIDGE (https://github.com/ebrahimabdelghfar/Carla_ASU_Bridge) [39], and the accompanying MPC and benchmarking packages, which are released together with the scenario definitions, the exported result CSVs and the figure and sweep generators at https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo/tree/feat-follow-premade-racing-track. The benchmarking packages compute one-step and multi-step prediction metrics online (RMSE, MAE, NRMSE, MaxAE, bias, standard deviation and R^2 by Welford's algorithm), compare estimated against ground-truth per-wheel lateral forces, and record controller-switching behaviour, and were built for this study.

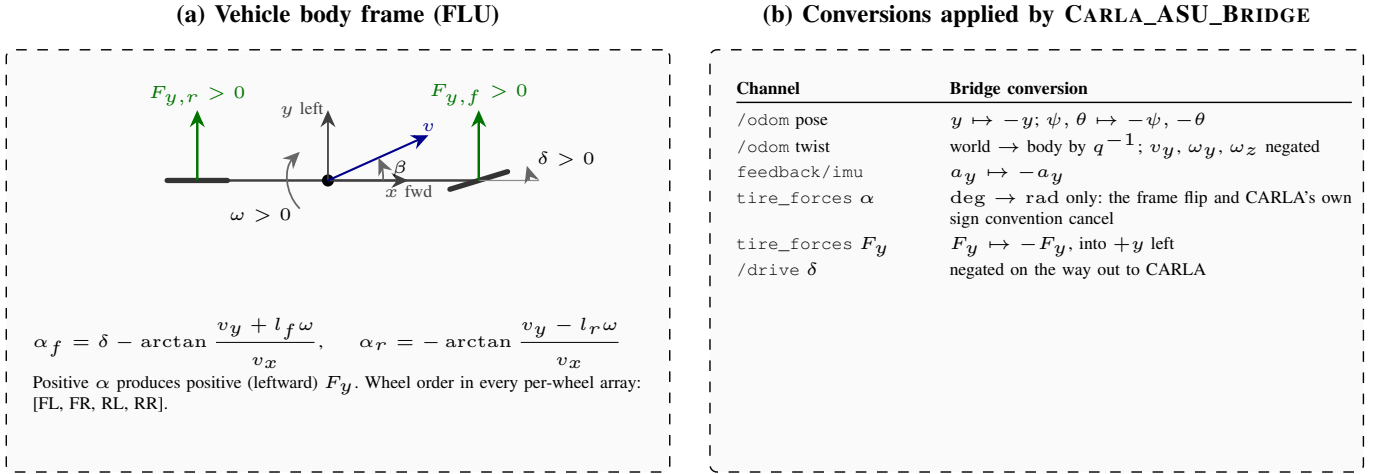


Fig. 9. Frame and sign conventions across the simulator–ROS 2 boundary. (a) All identification quantities are defined in the body frame: x forward, y left, ω and δ positive counter-clockwise, and slip angle and lateral force positive to the left, so that a positive α produces a positive F_y [54]. (b) The conversion applied per channel. The slip angle of the tire telemetry is the one exception that needs no sign change, because the frame flip and CARLA's own convention cancel.

ACKNOWLEDGMENT

The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.* [6], released under an open-source licence; this paper reports a port, a correction, and an extension of that work rather than a reimplementaion from scratch.

FUNDING

This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ETHICS

All results in this paper are obtained in simulation. No experiment involved human subjects, personal data, or the operation of a physical vehicle, so no human-subjects approval and no vehicle-safety protocol were required.

DATA AND CODE AVAILABILITY

The identification package, the ROS 2 bridge, the MPC and the benchmarking packages, the scenario definitions, the exported result CSVs behind every figure and table of Section VI, and the sweep generator behind Table IV are in the repositories listed in Appendix C; the top-level project repository is https://github.com/ebrahimabdelghfar/Ebrahim_Master_Thesis_repo/tree/feat-follow-premade-racing-track. ON-TRACK-SYSID is an extension of the ForzaETH release of [6] and is distributed under the MIT licence it declares in its package manifest; CARLA_ASU_BRIDGE is GPL-3.0. The two are used as separate ROS 2 overlays communicating only over ROS 2 topics and services, and neither links against the other, so no combined-work obligation arises between them; the identification and control packages in this work are released under the MIT terms the ON-TRACK-SYSID upstream requires.

REFERENCES

- [1] A. Liniger, A. Domahidi, and M. Morari, "Optimization-based autonomous racing of 1:43 scale RC cars," *Optimal Control Applications and Methods*, vol. 36, no. 5, pp. 628–647, 2015.
- [2] J. Kabzan, L. Hewing, A. Liniger, and M. N. Zeilinger, "Learning-based model predictive control for autonomous racing," *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 3363–3370, 2019.
- [3] N. Baumann, E. Ghignone, J. Kühne, N. Bastuck, J. Becker, N. Imholz, T. Kränzlin, T. Y. Lim, M. Lötscher, L. Schwarzenbach, L. Tognoni, C. Vogt, A. Carron, and M. Magno, "ForzaETH race stack—scaled autonomous head-to-head racing on fully commercial off-the-shelf hardware," *Journal of Field Robotics*, vol. 42, no. 4, pp. 1037–1079, 2024, preprint: arXiv:2403.11784.
- [4] H. B. Pacejka, *Tire and Vehicle Dynamics*, 3rd ed. Oxford, UK: Butterworth-Heinemann, 2012.
- [5] J. Kabzan, M. I. Valls, V. J. F. Reijgwart, H. F. C. Hendriks, C. Ehmke, M. Prajapat, A. Bühler, N. Gosala, M. Gupta, R. Sivanesan, A. Dhali, E. Chisari, N. Karnchanachari, S. Brits, M. Dangel, I. Sa, R. Dubé, A. Gawel, M. Pfeiffer, A. Liniger, J. Lygeros, and R. Siegwart, "AMZ driverless: The full autonomous racing system," *Journal of Field Robotics*, vol. 37, no. 7, pp. 1267–1294, 2020.
- [6] O. Dikici, E. Ghignone, C. Hu, N. Baumann, L. Xie, A. Carron, M. Magno, and M. Corno, "Learning-based on-track system identification for scaled autonomous racing in under a minute," *IEEE Robotics and Automation Letters*, vol. 10, no. 2, 2025, preprint: arXiv:2411.17508. Code: <https://github.com/ForzaETH/On-Track-SysID>.
- [7] L. Ljung, *System Identification: Theory for the User*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 1999.
- [8] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. 1st Annual Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 78. PMLR, 2017, pp. 1–16. [Online]. Available: <https://proceedings.mlr.press/v78/dosovitskiy17a.html>
- [9] E. Bakker, L. Nyborg, and H. B. Pacejka, "Tyre modelling for use in vehicle dynamics studies," SAE International, SAE Technical Paper 870421, 1987.
- [10] H. B. Pacejka and E. Bakker, "The magic formula tyre model," *Vehicle System Dynamics*, vol. 21, no. sup001, pp. 1–18, 1992.
- [11] F. Farroni, R. Lamberti, N. Mancinelli, and F. Timpone, "TRIP-ID: A tool for a smart and interactive identification of Magic Formula tyre model parameters from experimental data acquired on track or test rig," *Mechanical Systems and Signal Processing*, vol. 102, pp. 1–22, 2018.
- [12] T. A. Wenzel, K. J. Burnham, M. V. Blundell, and R. A. Williams, "Dual extended Kalman filter for vehicle state and parameter estimation," *Vehicle System Dynamics*, vol. 44, no. 2, pp. 153–171, 2006.
- [13] C. Sierra, E. Tseng, A. Jain, and H. Peng, "Cornering stiffness estimation based on vehicle lateral dynamics," *Vehicle System Dynamics*, vol. 44, no. sup1, pp. 24–38, 2006.

- [14] R. Rajamani, D. Piyabongkarn, J. Y. Lew, K. Yi, and G. Phnomchoeng, "Algorithms for real-time estimation of individual wheel tire-road friction coefficients," *IEEE/ASME Transactions on Mechatronics*, vol. 17, no. 6, pp. 1183–1195, 2012.
- [15] C. Ahn, H. Peng, and H. E. Tseng, "Robust estimation of road frictional coefficient," *IEEE Transactions on Control Systems Technology*, vol. 21, no. 1, pp. 1–13, 2013.
- [16] M. Acosta, S. Kanarachos, and M. Blundell, "Road friction virtual sensing: A review of estimation techniques with emphasis on low excitation approaches," *Applied Sciences*, vol. 7, no. 12, p. 1230, 2017.
- [17] S. Khaleghian, A. Emami, and S. Taheri, "A technical survey on tire-road friction estimation," *Friction*, vol. 5, no. 2, pp. 123–146, 2017.
- [18] Y.-H. J. Hsu, S. M. Laws, and J. C. Gerdes, "Estimation of tire slip angle and friction limits using steering torque," *IEEE Transactions on Control Systems Technology*, vol. 18, no. 4, pp. 896–907, 2010.
- [19] L. Hewing, A. Liniger, and M. N. Zeilinger, "Cautious NMPC with Gaussian process dynamics for autonomous miniature race cars," in *Proc. European Control Conference (ECC)*, 2018, pp. 1341–1348.
- [20] L. Hewing, J. Kabzan, and M. N. Zeilinger, "Cautious model predictive control using Gaussian process regression," *IEEE Transactions on Control Systems Technology*, vol. 28, no. 6, pp. 2736–2743, 2020.
- [21] J. Chrosniak, J. Ning, and M. Behl, "Deep dynamics: Vehicle dynamics modeling with a physics-constrained neural network for autonomous racing," *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5292–5297, 2024, preprint: arXiv:2312.04374.
- [22] Z. Wu, C. Hu, Y. Wang, L. Xie, and H. Su, "Vision-augmented on-track system identification for autonomous racing via attention-based priors and iterative neural correction," 2026. [Online]. Available: <https://arxiv.org/abs/2603.09399>
- [23] A. Gu, K. Goel, A. Gupta, and C. Ré, "On the parameterization and initialization of diagonal state space models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022, preprint: arXiv:2206.11893.
- [24] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proc. International Conference on Learning Representations (ICLR)*, 2022, preprint: arXiv:2111.00396.
- [25] E. Fiala, "Seitenkräfte am rollenden luftreifen," *VDI-Zeitschrift*, vol. 96, no. 29, pp. 973–979, 1954.
- [26] NVIDIA Corporation, "PhysX SDK 4.1 documentation: Vehicles — tire force computation (PxVehicleComputeTireForceDefault)," Software documentation, PhysX SDK 4.1, as bundled with Unreal Engine 4.26 and linked by CARLA 0.9.16, 2019, <https://github.com/NVIDIA-Omniverse/PhysX/tree/release/104.2/physx>.
- [27] M. Brunner, E. Ghignone, N. Baumann, and M. Magno, "R-CARLA: High-fidelity sensor simulations with interchangeable dynamics for autonomous racing," 2025. [Online]. Available: <https://arxiv.org/abs/2506.09629>
- [28] S. Sagmeister, P. Kounatidis, S. Goblirsch, and M. Lienkamp, "Analyzing the impact of simulation fidelity on the evaluation of autonomous driving motion control," 2026. [Online]. Available: <https://arxiv.org/abs/2602.07984>
- [29] J. Abdo, A. Shibu, M. Saeed, A. M. Aga, A. Sivaprazad, and M. Al-Musleh, "Simulating an autonomous system in CARLA using ROS 2," 2025. [Online]. Available: <https://arxiv.org/abs/2511.11310>
- [30] M. O'Kelly, H. Zheng, D. Karthik, and R. Mangharam, "FITENTH: An open-source evaluation environment for continuous control and reinforcement learning," in *Proc. NeurIPS 2019 Competition and Demonstration Track*, ser. Proceedings of Machine Learning Research, vol. 123. PMLR, 2020, pp. 77–89. [Online]. Available: <https://proceedings.mlr.press/v123/o-kelly20a.html>
- [31] R. Verschuere, G. Frison, D. Kouzoupis, J. Frey, N. van Duijken, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, 2022, preprint: arXiv:1910.13753.
- [32] G. Frison and M. Diehl, "HPIPM: A high-performance quadratic programming framework for model predictive control," *IFAC-PapersOnLine (Proc. 21st IFAC World Congress)*, vol. 53, no. 2, pp. 6563–6569, 2020.
- [33] A. Heilmeyer, A. Wischniewski, L. Hermansdorfer, J. Betz, M. Lienkamp, and B. Lohmann, "Minimum curvature trajectory planning and control for an autonomous race car," *Vehicle System Dynamics*, vol. 58, no. 10, pp. 1497–1527, 2020.
- [34] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. New York, NY, USA: Springer, 2012.
- [35] R. Bellman and K. J. Åström, "On structural identifiability," *Mathematical Biosciences*, vol. 7, no. 3, pp. 329–339, 1970.
- [36] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009.
- [37] K. J. Åström and B. Wittenmark, *Adaptive Control*, 2nd ed. Reading, MA, USA: Addison-Wesley, 1995.
- [38] P. Kaur, S. Taghavi, Z. Tian, and W. Shi, "A survey on simulators for testing self-driving cars," in *Proc. 4th Int. Conf. on Connected and Autonomous Driving (MetroCAD)*, 2021, pp. 62–70. [Online]. Available: <https://arxiv.org/abs/2101.05337>
- [39] E. Abdelghfar, "Formula AI simulator: A CARLA-based simulation bridge for autonomous formula student vehicles," Software, GPL-3.0, 2026, https://github.com/ebrahimabdelghfar/Carla_ASU_Bridge.
- [40] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.
- [41] A. Paszke, S. Gross, F. Massa, et al., "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [42] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
- [43] J. A. Nelder and R. Mead, "A simplex method for function minimization," *The Computer Journal*, vol. 7, no. 4, pp. 308–313, 1965.
- [44] R. Storn and K. Price, "Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, no. 4, pp. 341–359, 1997.
- [45] M. A. Branch, T. F. Coleman, and Y. Li, "A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems," *SIAM Journal on Scientific Computing*, vol. 21, no. 1, pp. 1–23, 1999.
- [46] P. Virtanen, R. Gommers, T. E. Oliphant, et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.
- [47] J. A. Cabrera, A. Ortiz, A. Simón, F. García, and A. Pérez La Blanca, "An alternative method to determine the magic tyre model parameters using genetic algorithms," *Vehicle System Dynamics*, vol. 41, no. 2, pp. 109–127, 2004.
- [48] A. Savitzky and M. J. E. Golay, "Smoothing and differentiation of data by simplified least squares procedures," *Analytical Chemistry*, vol. 36, no. 8, pp. 1627–1639, 1964.
- [49] J. Svendenius and M. Gäfvert, "A brush-model-based semi-empirical tire-model for combined slips," SAE International, SAE Technical Paper 2004-01-1064, 2004.
- [50] B. P. Welford, "Note on a method for calculating corrected sums of squares and products," *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962.
- [51] Institute of Automotive Technology, Technical University of Munich, "global_racetrajectory_optimization," Software, Institute of Automotive Technology, Technical University of Munich, 2019–2022, https://github.com/TUMFTM/global_racetrajectory_optimization.
- [52] G. C. Goodwin and R. L. Payne, *Dynamic System Identification: Experiment Design and Data Analysis*, ser. Mathematics in Science and Engineering. New York, NY, USA: Academic Press, 1977, vol. 136.
- [53] T. Foote and M. Purvis, "Standard units of measure and coordinate conventions," Open Source Robotics Foundation, ROS Enhancement Proposal 103, 2010, <https://www.ros.org/reps/rep-0103.html>.
- [54] SAE International, "Vehicle dynamics terminology," SAE International, Surface Vehicle Recommended Practice J670, 2008.